APPIEX ADMIN OS — COMPLETE FEATURE & SUB-FEATURE WORKFLOW BLUEPRINT

বাংলা-ইংরেজি Functional Design, Detailed Workflows, Screen Actions, Data, Permissions & Acceptance

Appiex | Internal Product Specification

20 October 2026

APPIEX ADMIN OS

Document contents

Part A: Foundation, roles, global metrics and shared experience

Part B: 12 Essential modules (M01–M12)

Part C: 9 Advanced modules (M13–M21)

Part D: 70 individually specified additional features (1–70)

Part E: Appiex Business Copilot

Part F: 12 integrated operational workflows

Part G: Data entities, events, API and integrations

Part H: Release plan, permissions and acceptance tests

Complete Feature-by-Feature Workflow Blueprint | Version **2.0** | **20** October **2026**

Website: https://appiex.com/
Product: Private agency operation system; proposed design document, not evidence that integrations/features already exist.
Audience: CEO, CMO, CTO, SDR, sales manager, growth, content, partnerships, finance, delivery, QA and engineering teams.
Scope: 12 essential modules + 9 advanced modules + 70 additional features + Business Copilot + cross-module workflows + data model + security/QA/release criteria.

Document reading guide

Screen/sub-feature = admin UI-তে কোন page, tab, filter, table, modal, drawer বা button থাকবে।

Workflow = trigger → validation → action/processing → approval → storage/sync → notification → reporting।

Data = কোন field/entity সংরক্ষিত হবে; Owner = authorized role; Status = অবস্থা ও transition।

Backend নেই এমন কাজ frontend-এ Demo / Not connected / Pending integration হিসেবে স্পষ্ট দেখাবে; real email/WhatsApp/payments/publication success fake করা যাবে না।

Real visitor সংখ্যা, Google ranking, impressions, paid invoices, bank balances, commission eligibility কোনোভাবেই fabricated হবে না। প্রতিটি external metric-এ provider, freshness timestamp, timezone, date range দেখাতে হবে।

Existing site-এর বাস্তব code/database/API এবং forms inspect করে path/field map reconcile করতে হবে; এই document নতুন target state।

Core business lifecycle

Traffic source (Google / paid / referral / social / direct) → website page view & consented event → inquiry/referral/partner application → spam & duplicate checking → CRM lead → SDR assignment → response/qualification → meeting → deal → proposal/contract → client onboarding → delivery milestones → verified invoice/payment → profitability / referral obligation → retention & reporting.

Appiex-এর পাঁচটি service taxonomy configurably initialize করবে: Digital Strategy; Web Development; UI/UX & 3D Design; AI Automation; SEO & Marketing। Referral commission-এর website copy up to 15%; এটি automatic fixed payout নয়—contract, eligibility, received payment ও approval-এর উপর নির্ভরশীল। Partner tracks: White Label, Affiliate, Collaboration।

PART A — FOUNDATION, USERS & GLOBAL EXPERIENCE

A1. Role-based access model

Super Admin — Typical read scope: Everything subject to verified ownership; Allowed operations: Governance, team and integration settings; Restricted operations: Sensitive action requires extra verification and logs

CEO/Executive — Typical read scope: Aggregated growth, sales, operations, finance; Allowed operations: Business approval, executive decisions; Restricted operations: Raw secrets/credentials

CMO/Growth — Typical read scope: Marketing, SEO, approved CRM summaries; Allowed operations: Campaign approval, CMS approval, experiments; Restricted operations: Payout execution, HR secrets

Sales Manager — Typical read scope: Pipeline, deals, sales messaging; Allowed operations: Routing, stage/quote approvals; Restricted operations: CMS publication unless granted

SDR — Typical read scope: Assigned leads, conversations, follow-up tasks; Allowed operations: Reply, notes, schedule, prepare quote drafts; Restricted operations: Org-wide data export, commission payment

Partnership Manager — Typical read scope: Partner/referral records; Allowed operations: Vetting, verification, agreement requests; Restricted operations: Final payout approval

Content Editor — Typical read scope: Blogs, projects, assets, drafts; Allowed operations: Edit, preview, submit approval; Restricted operations: Final publish when dual approval needed

Delivery/Project Manager — Typical read scope: Client-delivery objects; Allowed operations: Milestones, risks, task assignments; Restricted operations: Finance settlements

Finance — Typical read scope: Invoices, ledger, approved contracts; Allowed operations: Reconcile and approve payout stages; Restricted operations: Sales-message impersonation

Analyst/Auditor — Typical read scope: Authorized read-only dashboards and audit; Allowed operations: Export approved aggregate report; Restricted operations: Write or delete

Support — Typical read scope: Assigned support cases; Allowed operations: Response, assignment; Restricted operations: Unrelated client details

Client / Partner — Typical read scope: Own account/portal records; Allowed operations: Submit approvals, tickets, evidence; Restricted operations: Internal notes, margin, white-label identity

RBAC and ABAC দুটোই enforce করতে হবে: role + assigned record + field sensitivity + organization/project membership + action-specific permission। Frontend hidden button যথেষ্ট নয়; every API write must re-check authorization.

A2. Shared application shell and interactions

Sidebar groups: COMMAND, GROWTH, SALES, NETWORK, CONTENT, DELIVERY, FINANCE, ENGINEERING, INTELLIGENCE, SYSTEM। Collapse, favorites, recent pages, quick-create, role-aware counts।

Header: global search Ctrl/Cmd+K, date selector, compare period, timezone (default Asia/Dhaka), tenant/environment, profile, notification bell, connector health |

Every list: sortable/filterable paginated table, saved views, column chooser, export policy, bulk action review, owner/status chips, skeleton, error retry, empty state, drilldown drawer |

Every detail: history timeline, owner, notes, mentions, attachments, related entities, linked tasks, source/freshness, audit activity, next action |

Every destructive or external action: confirmation and reason; dangerous action may require approval/MFA; audit on backend.

Offline/unavailable connector: display Not connected, Permission expired, Delayed, Sync failed or Demo data, never represent provider results as zero.

A3. Metrics dictionary (definition must be consistent across pages)

Google Search clicks — Exact meaning: Clicks from Google Search results within selected Search Console property/date; Source: Google Search Console; Warning: Not equal to page CTA clicks

GA4 users — Exact meaning: Analytics-estimated users over selected period; Source: GA4; Warning: Subject to consent/modeling and provider processing

Page views — Exact meaning: Recorded page_view events; Source: GA4 / first-party events; Warning: May include repeats

Active now — Exact meaning: Provider-defined realtime activity; Source: GA4 Realtime / first-party tracker; Warning: Realtime may be delayed/estimated

CTA clicks — Exact meaning: Custom site click event with campaign/page/element metadata; Source: First-party event pipeline; Warning: Not proof of inquiry

Form submissions — Exact meaning: Server-accepted validated inquiry record; Source: Appiex API; Warning: Exclude spam/duplicates from qualified counts

Messages received — Exact meaning: Inbound message events ingested from connected channels; Source: Inbox API / authorized connectors; Warning: Email threading duplicates possible

Leads — Exact meaning: Unique CRM person/company opportunities per dedupe policy; Source: CRM; Warning: Unqualified leads aren't won deals

Won revenue — Exact meaning: Contracted value of successfully closed deals; Source: CRM/contracts; Warning: Not equivalent to collected money

Collected revenue — Exact meaning: Verified settled invoice payments, excluding refunds; Source: Finance ledger; Warning: Currency needs exchange-rate provenance

Referral commission due — Exact meaning: Eligible amount under configured written agreement after qualifying receipt; Source: Commission ledger; Warning: Not 15% of all leads

A4. Shared record conventions

All main records: immutable id, created/updated timestamp, created_by, source/origin, organization_id/tenant, owner, status, record version, audit reference, visibility class. IDs should be stable UUID/ULID. Monetary values: minor unit + currency, never binary float. PII encrypted at rest where appropriate; secrets in secret manager, not ordinary tables. Time stored UTC and rendered in user timezone. External IDs namespaced by provider.

PART B — 12 ESSENTIAL ADMIN MODULES (DETAILED)

M01. Executive Dashboard

কেন লাগবে: CEO যেন lead-to-revenue full picture ও আজকের বাধাগুলো দেখতে পারেন।

Screen/sub-feature inventory: Overview KPI grid; date/region/service filter; trends; acquisition-to-cash funnel; revenue by source; task queues; team workload; website/connector health; alerts; quick actions; saved executive views.

Step-by-step end-to-end workflow

Step **01** — Login → role/tenant permission resolve → default selected date/service/timezone এবং last saved view load।

Step **02** — Server KPI aggregator GA4/GSC/CRM/invoices/partner/service events fetch ও source health check করবে।

Step **03** — Cards: Visitors, Google clicks, CTA clicks, valid submissions, new/qualified leads, pending messages, meetings, won deals, collected revenue, referrals, partner applications, publish queue; প্রতিটিতে source/date/freshness।

Step **04** — KPI card click → corresponding filtered listing; funnel drop-off click → relevant segment; alerts click → action drawer।

Step **05** — Executive widget drag/reorder → private dashboard preferences save; approved CSV/PDF report export → report job+audit।

Step **06** — Important anomaly/overdue event → alert center; owner resolves → metric/queue updates on refresh।

Core data: kpi_snapshot, metric_definition, date_filter, user_dashboard_preference, report_job, connector_sync।

Status/state: Fresh/Delayed/Unavailable/Partial/Demo; personal saved view।

Permission ও approval: CEO/CMO see authorized aggregates; Finance sees financial detail; SDR only own metrics।

Failure/edge handling: Missing provider is — Not connected, never 0; multiple currencies need selected currency; incomplete time ranges marked।

KPI/acceptance: Widgets update filters in sync; exact filtered records reconcile with KPIs।

M02. Live Website Analytics

কেন লাগবে: Traffic source থেকে on-site behavior ও valid conversion বোঝা।

Screen/sub-feature inventory: Realtime overview; visitors now; geo/device/browser; landing pages; referrers/UTM; sessions/pages; CTA events; funnel; privacy-safe journey explorer; consent status; export.

Step-by-step end-to-end workflow

Step **01** — User opens Analytics → select today/7d/28d/custom + source/country/device/service।

Step **02** — Provider GA4 and consented first-party events read; normalize client_event_id, session and page metadata; preserve provider estimate flags।

Step **03** — Traffic timeline/chart + top landing pages; Realtime tab shows provider-defined active count; session-level explorer only if consent and role permit।

Step **04** — Click page → page-level visitor→CTA→form submission funnel; select CTA → events list with destination and campaign।

Step **05** — Segment builder saved view; compare previous period; privacy thresholds and retention limits applied before export।

Step **06** — Inbound lead links last-touch/original-source attribution with confidence status, no cross-device identity invented।

Core data: event_id, anonymous_session_id, consent_id, timestamp, path, referrer, utm_*, device, country, CTA_ID, linked_lead_id।

Status/state: Connected/Delayed/Consent-limited/Missing/Demo।

Permission ও approval: Growth/CEO, Analyst read; privileged technical access for raw events।

Failure/edge handling: Ad blockers and consent reduce tracking; page reload duplicates deduped; no session replay without consent and masking।

KPI/acceptance: Attribution join tested; CTA vs Google clicks clearly different।

M03. SEO Command Center

কেন লাগবে: Google organic visibility এবং technical SEO সমস্যাকে priority অনুযায়ী manage করা।

Screen/sub-feature inventory: SEO overview; Search Console query/page/country/device; keyword ranking samples; CTR opportunity; content decay; URL index state; sitemap; redirects; CWV; schema checks; competitor evidence; recommendations.

Step-by-step end-to-end workflow

Step **01 —** Authorized Google Search Console property connect (OAuth least scopes) and verify ownership; schedule safe incremental sync।

Step **02 —** Query/page records store date, device, country; display clicks/impressions/CTR/average position per chosen period।

Step **03 —** Opportunity engine flags high impressions & weak CTR, falling clicks, poor-performing pages, technical errors; attach evidence and confidence।

Step **04 —** User opens issue → affected URLs, baseline/last seen, suggested action, assignee, due date; send to CMS issue or engineering ticket।

Step **05 —** Editor prepares title/content change → approval → publish → wait for Google recrawl; monitor outcomes using pre/post comparable windows।

Step **06 —** Indexing/technical tabs show inspectable evidence and provider constraints; SEO reports never promise rank #1।

Core data: gsc_property, search_date, query, page, country, device, impressions, clicks, avg_position, issue, assigned_ticket।

Status/state: Detected/Triaged/In progress/Published/Monitoring/Resolved।

Permission ও approval: Growth owns; Content edits; Dev fixes technical; CMO publishes।

Failure/edge handling: GSC anonymizes some queries and delays data; no query volume invented; index status cannot be forced।

KPI/acceptance: GSC numbers match provider dates; issue-to-fix evidence traceable।

M04. Unified Inbox

কেন লাগবে: Every inbound business inquiry যেন response ownershipসহ traceable থাকে।

Screen/sub-feature inventory: All/Unassigned/Mine/Overdue tabs; channel chips; conversation list; thread pane; composer; templates; internal notes; attachments; assignment; SLA; customer summary; linked lead/deal; search.

Step-by-step end-to-end workflow

Step **01 —** First-party form, email, support, authorized WhatsApp connector থেকে normalized inbound event গ্রহণ; provider message ID দিয়ে dedupe।

Step **02 —** Match existing contact by trusted identifiers; unmatched → contact+lead suggestion; spam-risk quarantine queue।

Step **03 —** Rule engine priority/service/country/timezone অনুযায়ী sales/partner/support queue assign; SLA countdown শুরু।

Step **04 —** SDR opens thread → context, past contact history, form attachments; reply composer chooses allowed channel, template, optional AI draft।

Step **05 —** Send → permission/channel validity/rate/recipient check → connector send request → actual provider acknowledgment; then thread status Sent/Delivered/Failed update |

Step **06 —** Contact responds → reopen thread + notify owner; note and handoff preserve context; close requires resolution reason |

Core data**:** conversation, message, channel, provider_message_id, contact, assignee, attachments, SLA, send_status |

Status**/**state**:** New/Open/Pending customer/Pending internal/Resolved/Spam; Draft/Queued/Sent/Failed |

Permission **ও** approval**:** SDR assigned threads; managers all; finance private messages role-protected |

Failure**/**edge handling**:** Disconnected WhatsApp cannot be simulated as sent; attachment scanning; bounce/retry; duplicate ingestion |

KPI**/**acceptance**:** No accepted inquiry lost; first-response SLA auditable |

M05. CRM & SDR

কেন লাগবে: Visitor বা outreach lead থেকে contract conversion পর্যন্ত অপারেশন চালানো।

Screen**/**sub**-**feature inventory**:** Contacts, companies, lead queue, scoring, deal kanban, pipeline, tasks, meeting booking, proposals, documents, activity timeline, forecasts, SDR performance.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** Lead enters via form/referral/partner/manual/import/outreach → normalization + duplicate match + consent/source tracking |

Step **02 —** Lead owner assignment → SDR receives due task; status New → Contacting → Qualified / Disqualified with reason |

Step **03 —** SDR records requirement, service, budget, urgency, authority and next step; qualification score is explainable |

Step **04 —** Qualified lead → create opportunity/deal (source association retained); stages Discovery → Meeting → Proposal → Negotiation → Won/Lost |

Step **05 —** Meeting booking writes attendee/time/link and calendar provider sync; follow-up rules create tasks and reminders |

Step **06 —** Won deal triggers signed contract/client account/project onboarding; Lost records disqualification reason, compliant reactivation eligibility |

Core data**:** contact/company/lead/deal/service/budget/currency/stage/owner/next_action/consent/source, meetings/tasks/proposals |

Status**/**state**:** New/Contacting/Qualified/Disqualified; Discovery/Proposal/Negotiation/Won/Lost |

Permission **ও** approval**:** SDR assigned, Sales manager team, Finance approved prices |

Failure**/**edge handling**:** Duplicate companies; fake/spam entries; lead ownership conflicts; cross-currency reports; timezones |

KPI**/**acceptance**:** Every qualified lead has owner and next step; conversions reconcile |

M06. Referral Management

কেন লাগবে: Valid customer introductions verify এবং commission obligations safely track করা।

Screen**/**sub**-**feature inventory**:** Submission queue; referrer profile; referred company/client; consent/evidence; eligibility; dedupe; associated CRM lead/deal; commission rules; receipt linkage; payouts; disputes; referral links.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** refer form receives referrer and client details; secure storage of payout info separately; receipt/consent notice |

Step **02 —** Normalize person/company → duplicate/self-referral checks → Pending review; partner manager verifies genuine introduction with evidence |

Step **03 —** Valid referral generates/matches CRM lead, source referral ID remains permanent; rejected records keep reason and controlled retention |

Step **04 —** Deal closes Won + eligible contract + verified cash received → commission policy engine calculates allowed earned amount with cap, rate and exclusions |

Step **05 —** Finance checks ledger/evidence → independent approver confirms → payout queued; payment provider/bank reconciliation marks Paid |

Step **06 —** Referrer portal sees permitted status and reason, not confidential internal margin/client correspondence |

Core data**:** referral, referrer, referred_client, verification_evidence, policy_version, eligible_receipt, commission_ledger, payout |

Status**/**state**:** Submitted/Needs evidence/Verified/Rejected/Converted/Accrued/Approved/Paid/Disputed |

Permission ও approval**:** Partner manager verifies; Sales owns deal; Finance approves; dual approval for payouts |

Failure**/**edge handling**:** 15% is ceiling not default; duplicate or preexisting lead; refund/clawback; client confidentiality |

KPI**/**acceptance**:** Cannot pay twice; commission traceable to signed policy and received payment |

M07. Partnership Management

কেন লাগবে: White Label, Affiliate, Collaboration আলাদা lifecycle-এ পরিচালনা।

Screen**/**sub**-**feature inventory**:** Applications inbox; company/profile; model selector; screening; call scheduling; document/contract vault; partner scorecard; onboarding; referrals/projects; settlements; offboarding.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** /partners application ingested → type White Label/Affiliate/Collaboration + contact/company/interest captured |

Step **02 —** Reviewer requests missing documentation, screens fit/risk/conflict; assigns partner owner |

Step **03 —** Discovery call → contract terms including brand visibility, lead ownership, attribution, liability, commission and privacy |

Step **04 —** Authorized signer approves agreement; Approved partner gets scoped portal, referral assets, shared project workspace as contracted |

Step **05 —** Partner activities linked to referrals, projects, invoices and performance reports; renewal/termination workflow handles access revoke |

Step **06 —** Exceptional disputes/brand violations trigger hold; user-facing reports hide internal confidential data |

Core data**:** partner_application, organization, partnership_type, agreement, partner_user, activity, settlement |

Status**/**state**:** Applied/Screening/Call/Negotiating/Approved/Active/Suspended/Closed |

Permission ও approval**:** Partner manager + legal/commercial approver; finance payouts |

Failure**/**edge handling**:** Partner model change requires amended agreement; expired signed terms; white-label privacy |

KPI**/**acceptance**:** No portal access before agreement/approval; tracked obligations |

M08. Website CMS

কেন লাগবে: Live Appiex page content নিরাপদে বদলানো, deploy path-compatible publishing।

Screen/sub-feature inventory: Page list; block/section editor; header/footer menu; homepage hero; service sections; testimonials; FAQ; metadata; links; redirects; revision diff; staged preview; publish rollback.

Step-by-step end-to-end workflow

Step **01** — Editor selects editable page/section → base revision load; modify text/image/CTA/block order with validation।

Step **02** — CMS stores draft+diff, references stable media assets and content schema; preview shows actual responsive site components।

Step **03** — SEO/brand QA checks claims, alt text, broken links, mobile rendering, internal links, schema; unresolved blockers shown।

Step **04** — Submit approval → CMO/authorized approver reviews exact diff, rejects/comments or approves।

Step **05** — Publish job updates supported site content API/build hook → verified published URL/version; emit sitemap/revalidation event।

Step **06** — Failed build keeps current live version; rollback to prior revision needs authorization and audit।

Core data: page, page_revision, section_block, seo_metadata, asset, redirect_rule, publication_job।

Status/state: Draft/In review/Approved/Scheduled/Publishing/Published/Failed/Rolled back।

Permission ও approval: Editor drafts, CMO approves, engineering manages routes।

Failure/edge handling: Never publish CMS-only demo to live site; conflict merge; unsafe script sanitization।

KPI/acceptance: Preview matches live render; version rollback tested।

M09. Blog Management

কেন লাগবে: SEO-friendly editorial lifecycle সহ blog create ও publish করা।

Screen/sub-feature inventory: Blog table; rich editor; author; outline; slug; keywords; categories/tags; internal link helper; excerpt; hero image; schema; SEO preview; revisions; scheduler; analytics.

Step-by-step end-to-end workflow

Step **01** — Create post → choose author/service/target audience/intent; enter title, slug, excerpt, outline and body।

Step **02** — Autosave draft; insert approved images, alt text, citations and internal links; no fabricated case studies/results।

Step **03** — Validate schema, canonical, headings, originality and required SEO fields; preview desktop/mobile/social snippets।

Step **04** — Review/approve exact revision, schedule publication with explicit timezone or publish now।

Step **05** — Publishing adapter writes site CMS and rebuild/revalidate; record success only after verified response/URL।

Step **06** — Post-publication analytics link Search Console query/page and conversions; revisions maintain audit and rollback।

Core data: blog_post, body_blocks, author, category, slug, keyword_targets, media, approval, published_url।

Status/state: Draft/Review/Approved/Scheduled/Publishing/Live/Failed/Archived।

Permission ও approval: Content creates, CMO approves, only approved publisher triggers live।

Failure/edge handling: Duplicate slug, invalid canonical, missing required fields, scheduled timezone, provider failure।

KPI/acceptance: Live URL resolution checked; no auto-publish of unsafe AI text |

M10. Portfolio & Case Study Manager

কেন লাগবে: Appiex-এর কাজগুলো convincing কিন্তু client-approved উপায়ে publicly showcase করা।

Screen/sub-feature inventory: Projects list; project editor; services/capability; tech stack; media galleries; live/demo URL; outcomes with evidence; testimonials; privacy level; highlights/order; publication workflow.

Step-by-step end-to-end workflow

Step **01** — Project manager/editor creates internal project record or case study draft linked to client/delivery record |

Step **02** — Select visibility Public / Anonymized / Internal / White-label restricted; upload licensed media, brief, services, technologies |

Step **03** — Results/KPIs require evidence/provenance and client authorization; named client logos/testimonials need documented usage consent |

Step **04** — Responsive preview with image crop, alt text, project CTAs; SEO/brand QA and link check |

Step **05** — Approval chain: delivery lead validates facts → CMO validates copy → client permission if required → publish |

Step **06** — Case-study performance reports track views, inquiries and attribution; takedown request triggers unpublished revision |

Core data: case_study, client_permission, visibility, outcomes_evidence, media, category, technology, external_url, revision |

Status/state: Private/Draft/Pending client approval/Approved/Live/Withdrawn |

Permission ও approval: Project manager facts, client legal approval, marketer publication |

Failure/edge handling: Demo vs paid client clearly distinguished; white-label identity never exposed |

KPI/acceptance: No unapproved outcomes/names/images published |

M11. Team Management

কেন লাগবে: Staff, assignment, workload, role changes এবং access safety |

Screen/sub-feature inventory: Team directory; invitations; role matrix; teams; assignment rules; workload; calendar/availability; activity; leave/shift; access review; offboarding.

Step-by-step end-to-end workflow

Step **01** — Admin invites staff by verified email and selects base role/team/scopes; pending invitation expires |

Step **02** — Staff accepts → identity verification + MFA for privileged roles → team access provision |

Step **03** — Sales/partner/ticket assignment rules allocate work; workload dashboard shows overcapacity and unowned queues |

Step **04** — Manager can change team/task ownership with reason; project and thread history retained |

Step **05** — Access review checks stale permissions, dormant accounts, external invites and privileged actions |

Step **06** — Offboard → revoke sessions/tokens, transfer open objects, disable portal membership; preserve legally required audit |

Core data: user, team, role, permission_policy, invite, shift, assignment, access_log |

Status/state: Invited/Active/Suspended/Offboarded; On duty/Away |

Permission ও approval: Only Super Admin/HR authorized; managers assign within team |

Failure/edge handling: Conflicting permissions, last super-admin removal, employee departure, stale assignment |

KPI/acceptance: No orphaned records or lingering external access |

M12. Notification Center

কেন লাগবে: Time-sensitive actions ব্যক্তিকে এবং সঠিক channel-এ পৌঁছানো।

Screen/sub-feature inventory: All/Mentions/Assigned/Critical; unread; grouping; notification settings; routing rules; SLA alerts; quiet hours; digest; escalation; email/push/in-app.

Step-by-step end-to-end workflow

Step **01** — Business event generated (lead received, message unassigned, failed publish, payout approved) |

Step **02** — Notification engine evaluates recipient ACL, severity, dedupe key, quiet-hours and channel preferences |

Step **03** — In-app alert displays linked record and action; sensitive amounts/PII omitted from external previews |

Step **04** — User marks read/snoozes/resolves; critical SLA may escalate by organization policy to manager |

Step **05** — Notification delivery statuses, retries, failures and receipt logged; digest consolidates nonurgent alerts |

Core data: notification_rule, event_type, recipient, severity, payload_ref, delivery_attempt, read_status |

Status/state: Created/Queued/Delivered/Read/Snoozed/Escalated/Failed |

Permission ও approval: Recipient may configure self preferences; admin critical rules |

Failure/edge handling: Avoid duplicate alert storms; disable nonexistent channel; daylight saving/timezone |

KPI/acceptance: Important alerts delivered or actionable error surfaced |

PART C — 9 HIGH-VALUE ADVANCED MODULES

M13. AI Lead Intelligence

কেন লাগবে: SDR যেন evidence-ভিত্তিক lead prioritization পায়, কিন্তু AI-only rejection নয়।

Screen/sub-feature inventory: Lead scoring dashboard; score explanation; qualification factors; confidence; missing-data questionnaire; draft response; objection library; override reasons; calibration report.

Step-by-step end-to-end workflow

Step **01** — New lead validated → structured fields extracted from actual message with citation to message/field |

Step **02** — Deterministic fit rules + optional scored model evaluate budget fit, urgency, service, geography, authority and engagement; score explanation visible |

Step **03** — Low-confidence or missing fields flagged for manual qualification; scores never infer protected personal characteristics |

Step **04** — SDR reviews suggestions → manually adjusts qualified label/next task with reason; system records model version and override |

Step **05** — AI may draft reply or call agenda; human edits and approves before connector send |

Step **06** — Actual conversion later used to audit score usefulness, fairness, false positives and recalibrate |

Core data: lead_score, model_version, score_factors, confidence, human_override, draft, feedback |

Status/state: Unscored/Scored/Needs review/Overridden |

Permission ও approval: Sales Manager policy; SDR review; AI no send authority |

Failure/edge handling: Do not create ungrounded budget claims or silently reject leads |

KPI/acceptance: Explanation + provenance for all recommendations |

M14. Conversion Optimization Lab

কেন লাগবে: Data-driven landing-page improvement without misleading marketing claims |

Screen/sub-feature inventory: Heatmaps; scroll tracking; consented session replay; click goals; funnel comparison; form abandonment; experiments; hypothesis board; result interpretation.

Step-by-step end-to-end workflow

Step **01** — Mark measurable goal (valid service inquiry, meeting booked), target page/segment and privacy policy |

Step **02** — Implement event tracking with consent boundaries; validate goal events and sample size |

Step **03** — Hypothesis proposes CTA/header/form change; experiment editor creates controlled variants with stable assignment |

Step **04** — CMO authorizes traffic exposure; experiment starts only when implementation and QA pass |

Step **05** — Monitor conversion, bounce proxy and guardrail metrics; flag broken instrumentation before showing lift |

Step **06** — Analyst reviews uncertainty and duration; chooses Rollout/Pause/Discard; log experiment outcome; CMS publish approved variant |

Core data: experiment, variant, visitor_assignment, goal_event, traffic_exposure, consent, analysis_result |

Status/state: Draft/QA/Running/Paused/Completed/Rolled out |

Permission ও approval: CMO approval; Analyst reads; Dev implements |

Failure/edge handling: Cross-device contamination, low sample, tracking drop, sample ratio mismatch |

KPI/acceptance: Experiment conclusions distinguish observed lift vs statistical uncertainty |

M15. Proposal, Pricing & Contract Builder

কেন লাগবে: Sales quote থেকে verified agreement পর্যন্ত standardized workflow |

Screen/sub-feature inventory: Template library; service line items; multi-currency estimate; scope; timeline; assumptions; exclusions; milestones; taxes; approval; PDF; versioning; e-sign integrations.

Step-by-step end-to-end workflow

Step **01** — SDR opens qualified deal → New Proposal from template → pull verified client/project requirements |

Step **02** — Choose services, deliverables, payment milestones, assumptions, currency and pricing; optional margin calculation only authorized roles |

Step **03** — Preview branded PDF, expiry, scope/exclusion fields and client contact; Sales Manager approves discount/term thresholds |

Step **04** — Send only via connected authorized channel; provider confirmation stored; status Sent with timestamp |

Step **05 —** Client feedback creates revision; executed agreement verified using supported e-sign/legal process; attach signature reference |

Step **06 —** Accepted proposal triggers client onboarding and milestone invoice schedule; version history immutable |

Core data**:** proposal, revision, line_item, currency, milestone, signature_ref, approval |

Status**/**state**:** Draft/Approval/Sent/Viewed/Changes requested/Accepted/Expired/Rejected |

Permission **ও** approval**:** SDR prepare; manager approve; signing authority separate |

Failure**/**edge handling**:** Unsigned is not contracted; quoting taxes/currency errors; duplicate sends |

KPI**/**acceptance**:** Signed agreement matches approved version |

M16. Profitability & Revenue Intelligence

কেন লাগবে: Collected cash, cost, committed work ও forecast আলাদা রাখে।

Screen**/**sub**-**feature inventory**:** Revenue KPI, AR aging, per-service profit, cost allocation, currency conversion, contract value, margin trend, refund, expense approvals, forecast scenarios.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** Closed-won deal value and signed agreement ingest into booked revenue; invoice separately issued |

Step **02 —** Finance reconciles bank/payment records → verified receipts mark cash collected, not from manual UI guess |

Step **03 —** Cost/time/travel/vendor expenses allocated to delivery/project by cost center |

Step **04 —** Ledger calculates project gross margin, contribution and currency-adjusted totals with rate/date shown |

Step **05 —** Forecast scenario uses explicitly labeled assumptions; does not overwrite ledger |

Step **06 —** CEO drilldowns identify high-margin services, overdue invoices, concentrated revenue and negative-margin jobs |

Core data**:** contract, invoice, payment, credit_note, exchange_rate, vendor_cost, timesheet, allocation |

Status**/**state**:** Booked/Invoiced/Part-paid/Paid/Overdue/Refunded |

Permission **ও** approval**:** Finance owns reconciled receipts; CEO read; PM sees scoped cost |

Failure**/**edge handling**:** Double-count, VAT inclusion, refund, exchange-rate changes |

KPI**/**acceptance**:** Revenue and cash never conflated; ledger totals reconcile |

M17. Marketing Campaign Center

কেন লাগবে: Cross-channel campaigns ও revenue attribution পরিচালনা।

Screen**/**sub**-**feature inventory**:** Campaign list; creative; channel links; UTM builder; budget; spend; status; ad connector health; leads by campaign; CAC; ROAS; approvals.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** CMO creates campaign with name, objective, country, segment, channel, budget, UTM naming convention |

Step **02 —** Media/creative draft attaches approved website page or campaign asset; approval gate checks brand and claims |

Step **03 —** Optional platform connector returns campaign status/spend; disconnected channels remain Not connected |

Step **04 —** First-party leads capture campaign UTMs; attribution engine links spend→lead→opportunity→verified sale with model label।

Step **05 —** CMO monitors spend vs qualified leads, CAC and collected revenue, pauses or revises according to approved process।

Step **06 —** Export campaign evidence and summarized learning; honor platform scopes/rate limits।

Core data**:** campaign, creative, utm, spend_observation, attributed_lead, provider_ad_id।

Status**/**state**:** Draft/Approval/Running/Paused/Ended/Sync error।

Permission **ও** approval**:** Growth admin/CMO; Finance spend verification।

Failure**/**edge handling**:** Ad click mismatch and tracking loss; platform reporting differences।

KPI**/**acceptance**:** No phantom spend or zero metrics for disconnected connectors।

M18. Content Calendar & Social Publishing

কেন লাগবে: Blog/social/portfolio/SEO planning একসাথে এবং branded review।

Screen**/**sub**-**feature inventory**:** Monthly/week calendar; content cards; campaign linkage; channels; approval chain; assets; scheduling; copy variants; publishing jobs; performance.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** Content manager creates content request from service/marketing goal; assign owner, due date, channel, keywords।

Step **02 —** Draft text/media → brand check, accessibility, legal/client-consent check → internal review।

Step **03 —** Editor schedules per channel/timezone; scheduler validates provider account permission and asset compatibility।

Step **04 —** Publish via supported API or create manual posting task for unsupported destinations; store actual post URL only if confirmed।

Step **05 —** Failure retries with idempotency and alert; published post metrics pulled where authorized।

Step **06 —** Repurpose approved case study into social/blog variants without unapproved confidential details।

Core data**:** content_item, calendar_slot, channel, copy_variant, media, approval, publish_job।

Status**/**state**:** Idea/Draft/Review/Approved/Scheduled/Publishing/Published/Failed।

Permission **ও** approval**:** Editors draft, CMO approve, publisher execute।

Failure**/**edge handling**:** Rate limits, scheduled provider offline, deleted post, consent revoked।

KPI**/**acceptance**:** All scheduled items show exact destination and real delivery status।

M19. Website Health & Quality Control

কেন লাগবে: Growth এবং Engineering যেন website quality issues দ্রুত catch করে।

Screen**/**sub**-**feature inventory**:** Uptime; synthetic form tests; broken links; SEO metadata scanner; CWV trends; deploy releases; incident board; UI regression; redirects; accessibility checks.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** Scheduled checks run public routes, sitemap, SSL/domain and first-party form health (synthetic submission separated from actual leads)।

Step **02 —** Collect lab performance and permitted field metrics with environment labels; compare thresholds।

Step **03 —** Failure creates triaged issue with URL, reproduction, severity, last passing time, owner।

Step **04 —** Developer links code fix/deployment; QA runs validations after release; verify public route recovers।

Step **05 —** Incidents closed with root cause, downtime estimate grounded in monitoring, preventative task; notify client if contract dictates।

Core data**:** health_check, monitor, alert, issue, deployment_ref, incident, verification।

Status**/**state**:** Healthy/Warning/Down/Investigating/Fixed/Monitoring।

Permission **ও** approval**:** Engineering owns; Growth views SEO; CEO summary।

Failure**/**edge handling**:** False downtime due to probe network; tests should not submit public spam।

KPI**/**acceptance**:** Auditable first detection→resolution timestamps।

M20. Automation Control Center

কেন লাগবে: n8n এবং internal workflows visible, safe, retryable করা।

Screen**/**sub**-**feature inventory**:** Workflow catalog; trigger list; owners; execution timeline; parameters; secrets references; dry-run; error queue; retry; quota; alert rules; versioning.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** Admin connects n8n with scoped API token securely; read workflows, active state and executions।

Step **02 —** Create automation intent (new lead→assign/notify, overdue task→remind, approved content→schedule) with policy-approved actions।

Step **03 —** Dry-run on synthetic event shows expected steps; sensitive external action requires review and approval।

Step **04 —** Enable → every trigger gets idempotency key; execution logs show step statuses, duration, retry and sanitized errors।

Step **05 —** Failure routes to dead-letter/retry queue; operations can replay safe execution or disable rule।

Step **06 —** Tokens rotated and execution history retained per policy; never expose secrets in frontend logs।

Core data**:** workflow, trigger, execution, idempotency_key, integration_ref, retry_policy, dlq।

Status**/**state**:** Draft/Tested/Enabled/Paused/Failed; Run queued/running/succeeded/failed।

Permission **ও** approval**:** Ops/Engineering manages; business owners approve action scope।

Failure**/**edge handling**:** Webhook repeat, loop, upstream outage, revoked token।

KPI**/**acceptance**:** Exactly-once effects where feasible; duplicate messages prevented।

M21. Client & Partner Portals

কেন লাগবে: External users যেন only authorized shared record দেখতে ও action নিতে পারেন।

Screen**/**sub**-**feature inventory**:** Client login; project tracker; assets; milestone approvals; invoices; helpdesk; meeting notes; partner referrals; commission status; agreement docs.

Step**-**by**-**step end**-**to**-**end workflow

Step **01 —** CRM Won + verified contact triggers invitation, not automatic full access; access tied to one organization/project |

Step **02 —** Client signs in → only owned projects, deliverables and invoices; internal cost/notes hidden |

Step **03 —** Client uploads required assets → malware scan + permission; marks feedback/approval with timestamp and version |

Step **04 —** Partner accesses own referral status and commission statement; other partners' leads not visible |

Step **05 —** Contract expires/offboarding → revoke access and preserve required records under retention policy |

Core data**:** portal_user, membership, project_share, approval, shared_file, partner_referral_scope |

Status**/**state**:** Invited/Active/Revoked; Pending review/Approved/Changes requested |

Permission ও approval**:** Client/Partner own data; staff delegates scoped access |

Failure**/**edge handling**:** White-label branding, wrong tenant access, shared link expiry |

KPI**/**acceptance**:** Cross-tenant access tests pass for every API |

PART D — 70 ADDITIONAL FEATURES: EACH WITH UI, SUB-FEATURES & WORKFLOW

নিচের 70টি আইটেম মূল module-এর extensions। প্রতিটি আলাদাভাবে implementable, এবং সংশ্লিষ্ট মূল module-এর rules inherit করবে। 41–45 Engineering/Infrastructure items-ও অন্তর্ভুক্ত করা হয়েছে, যাতে numbering-এ কোনো বাদ না পড়ে।

D1. Growth, Traffic & Visitor Intelligence (1–10)

1. Visitor Journey Explorer

কাজ/উদ্দেশ্য: একই consented session-এর landing থেকে conversion journey map |

Screen ও sub**-**features**:** Journey timeline; page/event sequence; landing/source filter; funnel stage; journey compare; PII mask; retention settings |

পূর্ণ workflow**:** Visitor consent + identifier → page/events collect → event-time order ও dedupe → stages display → linked verified lead থাকলে attribution annotate → analyst filters drop-off and creates optimization task |

Data**/**connection**:** session_id, consent_scope, page_path, event_name, event_time, campaign, lead_link

Business rule**,** access ও validation**:** No silent identity stitching across devices; anonymize IP and enforce minimum group thresholds Owner**:** Growth Analyst |

Exceptions**/**empty state**:** Consent missing → aggregated-only; events incomplete → Partial tracking badge

2. Traffic Anomaly Detector

কাজ/উদ্দেশ্য: অস্বাভাবিক traffic/spam/measurement outage আলাদা করে warning |

Screen ও sub**-**features**:** Baseline chart; configurable deviation thresholds; hour/day segmentation; known campaign calendar; alert severity; incident acknowledgement |

পূর্ণ workflow**:** Historical baseline train/compute → periodic traffic observation → seasonality/campaign adjustment → sustained anomaly detect → source health/spam check → notify owner → investigation case open → mark cause/resolved |

Data/connection: metric_series, baseline, alert_threshold, anomaly_window, incident_ref

Business rule, access ও validation: Low-volume noisy series avoid false emergencies; compare like-for-like windows Owner: Growth/Engineering।

Exceptions/empty state: GA4 outage বা tracking change হলে Instrumentation suspect

3. Geographic Opportunity Map

কাজ/উদ্দেশ্য: কোন দেশ থেকে profitable qualified leads আসছে দেখা।

Screen ও sub-features: Country map/table; city where privacy permits; clicks→leads→won→cash; service breakdown; comparison; export।

পূর্ণ workflow: Provider region aggregate → CRM lead origin/country join → currency normalize → qualification/revenue per area → segment drilldown → export approved market shortlist।

Data/connection: country, region, visitor_count, lead_count, pipeline_value, collected_revenue

Business rule, access ও validation: Never infer precise address or reveal single-user location; small counts suppressed Owner: CMO/CEO।

Exceptions/empty state: Unknown geos separate bucket; VPN/attribution uncertainty labeled

4. Visitor Segmentation

কাজ/উদ্দেশ্য: High-intent traffic cohorts বুঝে marketing optimize করা।

Screen ও sub-features: New/returning selector; channel/source; device; campaign; engagement; saved segments; cohort comparison।

পূর্ণ workflow: Define segment rules → validate definitions → query privacy-eligible events → calculate volume, engagement, conversion → compare segments → save audience definition for reporting/experiment (not unsolicited targeting)।

Data/connection: segment_definition, event_conditions, date_window, cohort_aggregate

Business rule, access ও validation: Consent/marketing opt-in required for identifiable outreach Owner: Growth।

Exceptions/empty state: Sample size low → suppress precision; cross-device unresolved

5. Form Abandonment Analytics

কাজ/উদ্দেশ্য: Inquiry forms কোথায় user drop করছে।

Screen ও sub-features: Form start/completion chart; field error heatmap; step durations; mobile split; funnel; troubleshooting notes।

পূর্ণ workflow: Instrument form_start, field validation error and form_submit_success (never actual typed contents) → aggregate stages → show abandonment percentages → identify issue → create experiment/bug → measure change।

Data/connection: form_id, form_step, error_code, event_time, device, conversion_id

Business rule, access ও validation: Never record passwords, message body, card/payout details or keystrokes Owner: Growth/Product।

Exceptions/empty state: Consent denied or JS blocked → coverage note; spam excluded

6. Session Replay

কাজ/উদ্দেশ্য: UI usability issue troubleshoot with privacy।

Screen ও sub-features: Session list; consent proof; replay controls; mask regions; retention and delete; feedback/issue link।

পূর্ণ workflow**:** Obtained explicit permitted consent → recording SDK masks text, input, attachments and PII → store encrypted limited TTL → authorized analyst view → clip issue note without exporting sensitive content → delete on request/expiry।

Data/connection**:** replay_id, consent_version, masked_payload, TTL, access_audit

Business rule**,** access ও validation**:** Disable recording by default for finance/portal/private/admin screens Owner**:** Approved Growth/QA।

Exceptions/empty state**:** Masking failure → suspend capture and purge compromised replay

7. Scroll & Click Heatmaps

কাজ/উদ্দেশ্য: কোন UI section seen/clicked সবচেয়ে বেশি।

Screen ও sub-features**:** Page selector; device viewport; scroll depth; aggregated click overlay; CTA label; period compare; sample counts।

পূর্ণ workflow**:** Collect consented scroll/click coordinates with responsive layout version → aggregate by breakpoint/section → suppress sparse bins → visualize on version-matched screenshot → propose layout test।

Data/connection**:** page_revision, viewport, element_id, x/y normalized, scroll_depth

Business rule**,** access ও validation**:** Exclude sensitive elements and individual journeys without consent Owner**:** UX/Growth।

Exceptions/empty state**:** Page redesign causes old coordinates mismatch; mark incompatible

8. Customer Acquisition Attribution

কাজ/উদ্দেশ্য: Visit থেকে earned revenue পর্যন্ত source explain করা।

Screen ও sub-features**:** First/last touch; campaign path; channel comparison; unattributed bucket; join confidence; revenue drilldown।

পূর্ণ workflow**:** Capture lawful first-party source UTMs → form submits attribution cookie/session ref → link contact→deal→contract→receipt → assign credit using stated model → CEO compares qualified lead and collected revenue → export evidence।

Data/connection**:** attribution_touch, lead_id, deal_id, invoice_id, payment_id, model

Business rule**,** access ও validation**:** No probabilistic individual identity without justification; clearly show untracked portion Owner**:** Growth/Finance।

Exceptions/empty state**:** Missing tracking ≠ Direct; duplicate leads excluded from conversion

9. Conversion Path Comparison

কাজ/উদ্দেশ্য: Converted vs abandoned cohorts-এর sequence পার্থক্য।

Screen ও sub-features**:** Journey path visualization; winning/lost cohort; channel and page filters; segment size; hypotheses task।

পূর্ণ workflow**:** Create valid conversion definition → select consent-safe cohorts → build aggregate ordered event paths → calculate typical route/drop-offs → compare fair traffic segments → add hypothesis to experiment backlog।

Data/connection**:** conversion_goal, path_edges, cohort_size, date_range

Business rule**,** access ও validation**:** Avoid correlation-as-causation; protected/small cohorts suppressed Owner**:** Growth Analyst।

Exceptions/empty state**:** Insufficient conversion count → no misleading graph

10. Website Experiment Manager

কাজ/উদ্দেশ্য: Test hypothesis deploy, measure, decide।

Screen ও sub-features: Experiment board; goals; variants; audience split; QA checklist; guardrail metric; significance guidance; rollout log।

পূর্ণ workflow: Hypothesis → measurable goal & stable assignment → create variants → engineering QA → CMO approve → traffic split → event QA and monitoring → result review → rollout or revert → archive learnings।

Data/connection: experiment_id, variant_id, goal_id, allocation, assignment_key, result

Business rule, access ও validation: No mid-test selective winner declaration; keep allocation/cross-device assumptions explicit Owner: CMO/Product।

Exceptions/empty state: Sample ratio mismatch or page bug auto-pauses test

D2. SEO and AI Search Intelligence (11–20)

11. SEO Opportunity Finder

কাজ/উদ্দেশ্য: High-impression, low-CTR queries priority assign করা।

Screen ও sub-features: Opportunity table; query/page pairs; actual GSC baselines; opportunity scoring; edit task; after-change evaluation।

পূর্ণ workflow: GSC sync → minimum-impression filter → compare CTR with suitable page/query baseline → inspect intent/snippet → propose fix → approve task → publish change → monitor comparable periods।

Data/connection: query, page, clicks, impressions, ctr, position, prior_window

Business rule, access ও validation: Opportunity is hypothesis, not traffic guarantee Owner: SEO Lead।

Exceptions/empty state: Sparse/anonymized GSC rows → mark estimate/unknown

12. Keyword Cannibalization Detector

কাজ/উদ্দেশ্য: একই intent-এর জন্য একাধিক URL conflict খোঁজা।

Screen ও sub-features: Query-to-pages matrix; overlapping ranking URLs; search intent notes; recommended canonical/merge/link strategies।

পূর্ণ workflow: Find shared query across pages → assess whether distinct intent → detect harmful overlap using impression/click trends → SEO reviews → choose content differentiation/merge/canonical → publish → monitor।

Data/connection: gsc_query_page, url, intent_tag, canonical_url, issue

Business rule, access ও validation: Do not canonical distinct valuable pages automatically Owner: SEO/Content।

Exceptions/empty state: Overlaps for legitimate product variants not necessarily problems

13. Content Decay Monitor

কাজ/উদ্দেশ্য: আগে ভালো চলা content decline flag।

Screen ও sub-features: Trend plots; decay list; comparable windows; updated date; competitors; refresh task and outcomes।

পূর্ণ workflow**:** Compute rolling click/impression deltas for indexed pages → exclude seasonal/technical outages → classify content freshness issue → editor audit and revise → approve/publish → 4–8-week follow-up।

Data**/**connection**:** page_id, gsc_clicks_history, last_content_update, decay_score

Business rule**,** access **ও** validation**:** Do not label short noise as decline Owner**:** SEO Lead।

Exceptions**/**empty state**:** Property move / migration creates false decline → rebase

14. Internal Link Intelligence

কাজ/উদ্দেশ্য: Topical authority এবং crawl paths improve।

Screen **ও** sub**-**features**:** Link graph; orphan pages; anchor suggestions; broken internal link; priority link tasks।

পূর্ণ workflow**:** Crawl public site and CMS drafts → build internal link graph → identify relevant contextual link with topical relevance → editor reviews anchor/target → CMS edit → post-publish crawl verify।

Data**/**connection**:** source_url, target_url, anchor, link_status, content_tags

Business rule**,** access **ও** validation**:** Never auto-insert spammy exact-match anchors Owner**:** SEO/Content।

Exceptions**/**empty state**:** JS-generated links may need rendered crawler verification

15. Competitor Content Gap

কাজ/উদ্দেশ্য: Competitor-covered useful search intents identify করা।

Screen **ও** sub**-**features**:** Approved competitor list; topical gap report; SERP source links; content brief; prioritization।

পূর্ণ workflow**:** Define comparable legitimate competitors → collect permitted public page/query evidence → cluster topics → exclude irrelevant keywords → estimate fit and effort → brief article/service page → human validation।

Data**/**connection**:** competitor_domain, source_url, topic_cluster, match_score

Business rule**,** access **ও** validation**:** Public evidence only; no unverified traffic estimates labeled factual Owner**:** CMO/SEO।

Exceptions**/**empty state**:** Thin / outdated source → evidence age warning

16. Schema Markup Manager

কাজ/উদ্দেশ্য: Structured data safely manage ও validate করা।

Screen **ও** sub**-**features**:** Schema types; JSON-LD form; page map; validation; preview; deployment tracking; errors।

পূর্ণ workflow**:** Choose suitable schema based on real page info → editor enters verified claims → generate JSON-LD → validate schema syntax/eligibility → approve → deploy site build → check rendered HTML and warnings।

Data**/**connection**:** schema_type, entity_ref, json_ld, validation_result, published_revision

Business rule**,** access **ও** validation**:** No fake reviews, ratings, founders, awards or fabricated structured data Owner**:** SEO/Developer।

Exceptions**/**empty state**:** Schema valid ≠ rich result guaranteed; report eligibility separately

17. Search Intent Classifier

কাজ/উদ্দেশ্য: Keyword intent অনুযায়ী content planning।

Screen **ও** sub**-**features**:** Keyword clustering; intent labels; confidence; manual override; page mapping; content briefs।

পূর্ণ workflow: Import actual GSC/research keywords → classify informational/commercial/transactional/navigational → inspect ambiguous terms → map to current/planned page → identify content gaps → approve plan।

Data/connection: keyword, label, confidence, target_page, reviewer

Business rule, access ও validation: Human correction persisted; multilingual Bangla/Banglish classification review Owner: SEO/Content।

Exceptions/empty state: Brand homonyms and transliterated words uncertain

18. AI Search Visibility Tracker

কাজ/উদ্দেশ্য: Selected generative search products-এ documented citation/mention monitoring।

Screen ও sub-features: Tracked prompts; jurisdiction/device/model; captured response; cited links; mention classification; sampling caveat।

পূর্ণ workflow: Create neutral fixed prompt set → authorized/manual repeatable sampling on stated dates → save source output with model/location/time → detect brand mention and citation → trend as sampled observation only।

Data/connection: prompt_set, provider, run_time, location, answer_snapshot, citation_url, mention

Business rule, access ও validation: No claim of exhaustive ChatGPT/Google AI visibility or hidden ranking data Owner: SEO Research।

Exceptions/empty state: Model stochastic responses; API availability and terms restrictions labeled

19. Backlink Opportunity Pipeline

কাজ/উদ্দেশ্য: High-quality editorial relationships track, not spam।

Screen ও sub-features: Prospect database; relevance; contact consent; outreach stage; earned link check; nofollow/sponsored labels।

পূর্ণ workflow: Discover relevant publication/public resource → manual qualification → pitch value-driven asset through lawful channel → follow-up limit → verify public link and context → measure referral traffic।

Data/connection: prospect_site, relevance, contact, outreach_history, backlink_url, rel_type

Business rule, access ও validation: No link buying, mass spam, or automatic fabricated testimonials Owner: SEO Outreach।

Exceptions/empty state: Site rejects proposal → stop; broken link → follow-up task

20. Search Indexing Monitor

কাজ/উদ্দেশ্য: Important URLs discovery/indexing problems detect।

Screen ও sub-features: Sitemap list; canonical audit; robots checks; HTTP status; URL inspections; issue queue।

পূর্ণ workflow: Scheduled public crawl+GSC available reports → identify noindex, redirect chains, 404, sitemap mismatch → triage priority → engineer fixes → resubmit sitemap if appropriate → monitor evidence।

Data/connection: url, crawl_state, index_evidence, canonical, robots_rule, last_checked

Business rule, access ও validation: Google decides indexing; submission is not guarantee Owner: SEO/Engineering।

Exceptions/empty state: Inspection quota exceeded → stale notice instead of false result

D3. SDR & Sales Intelligence (21–30)

21. Ideal Client Profile (ICP) Manager

কাজ/উদ্দেশ্য: Target vertical/country অনুযায়ী lead qualification ও prospecting optimize।

Screen ও sub-features: ICP library; country/sector/company size/budget; exclusion; service match; fit explanations; segment KPI।

পূর্ণ workflow: Sales Manager drafts ICP → define criteria from past qualified wins and strategy → approve segment → new leads receive transparent fit score → SDR filters queue by ICP → compare win rates and revise quarterly।

Data/connection: icp_id, sector, region, budget_band, persona, disqualifiers, version

Business rule, access ও validation: Scoring not based on protected traits; avoid blanket geo exclusion without business reason Owner: Sales Manager/CMO।

Exceptions/empty state: Small historical sample → label heuristic, not predictive

22. Sales Sequence Builder

কাজ/উদ্দেশ্য: Consistent compliant multi-step follow-up।

Screen ও sub-features: Email templates; steps; wait periods; business hours; personalization; unsubscribe; stop-on-reply; sequence preview।

পূর্ণ workflow: Create sales sequence → choose eligible opted-in/lawful audience and steps → manager approve templates/cadence → SDR enrolls lead → queue sends with rate limits → inbound reply/opt-out automatically stops sequence → activity logged।

Data/connection: sequence, steps, enrolled_contact, send_job, reply_signal, unsubscribe

Business rule, access ও validation: Honor consent/legitimate business outreach law and provider terms; do not send bulk unsolicited spam Owner: Sales Ops।

Exceptions/empty state: Bounce, wrong timezone, duplicate enrollments → pause and notify

23. Lead Response SLA

কাজ/উদ্দেশ্য: প্রথম reply এবং resolution delay visible করা।

Screen ও sub-features: Queue timers; response business hours; owner scoreboard; overdue alerts; escalation history।

পূর্ণ workflow: Inbound accepted → SLA policy chosen by source/priority/business hours → owner assigned → clock calculated → warning at threshold → manager escalate if missed → genuine outbound reply closes first-response timer।

Data/connection: sla_policy, inbound_received_at, first_response_at, assignee, escalation

Business rule, access ও validation: Internal note does not count as customer reply; handoff retains original start time Owner: Sales Manager।

Exceptions/empty state: Unassigned, weekends/timezones, connector send failure count as not responded

24. Lost Deal Analysis

কাজ/উদ্দেশ্য: Sales failure থেকে systematic lessons।

Screen ও sub-features: Lost reason taxonomy; competitor field; pricing objection; timeline; lost notes; win/loss dashboard।

পূর্ণ workflow**:** Deal moves to Lost → reason required → SDR selects coded cause + notes/evidence → manager reviews important loss → monthly trend by ICP/channel/service → create pricing/portfolio/process improvement task।

Data**/**connection**:** deal_id, lost_reason, competitor, quoted_value, stage_died, feedback

Business rule**,** access **ও** validation**:** Avoid speculative motive claims; allow unknown rather than forced false reason Owner**:** Sales Manager।

Exceptions**/**empty state**:** Duplicate deal or no proposal → stage-specific required fields

25. AI Sales Conversation Coach

কাজ/উদ্দেশ্য: SDR response quality ও discovery improve।

Screen **ও** sub**-**features**:** Conversation summary; extracted requirements; recommended questions; objections; coaching rubric; feedback controls।

পূর্ণ workflow**:** Authorized conversation ingested → transcription/summary with permission → model returns suggestions citing actual messages → SDR sees and edits → manager calibrates coaching quality → record feedback without hidden employee profiling।

Data**/**connection**:** thread_id, cited_message_ids, coach_version, advice, feedback

Business rule**,** access **ও** validation**:** Human review before reply; do not infer customer sensitive traits Owner**:** SDR/Sales Lead।

Exceptions**/**empty state**:** No consent/transcription rights → disable recording/analysis

26. Lead Enrichment

কাজ/উদ্দেশ্য: Verified company-level context যোগ করা।

Screen **ও** sub**-**features**:** Company lookup; websites; sector; size range; verification/confidence; data source; merge review।

পূর্ণ workflow**:** Lead company/domain provided → approved data provider lookup → show source + freshness → human selects mergeable facts → save verified fields → recalculate fit score।

Data**/**connection**:** provider_source, company_domain, employee_range, industry, last_verified

Business rule**,** access **ও** validation**:** Do not scrape restricted personal data or silently overwrite first-party fields Owner**:** Sales Ops।

Exceptions**/**empty state**:** Conflicting companies with same name → require manual match

27. Sales Forecasting

কাজ/উদ্দেশ্য: Pipeline weighted forecast ও target gap বোঝা।

Screen **ও** sub**-**features**:** Forecast buckets; stage probability; committed/best-case; close date; actual vs expected; scenario assumptions।

পূর্ণ workflow**:** Load open deals and history → stage conversion probabilities from reliable outcomes or manual assumptions → weighted amount by close window/currency → show projected vs actual collected separately → manager adjusts labeled scenarios।

Data**/**connection**:** deal_value, stage, expected_close, probability, forecast_run

Business rule**,** access **ও** validation**:** Future forecast not booked or collected revenue; preserve model assumptions Owner**:** CEO/Sales Manager।

Exceptions**/**empty state**:** Low sample probability → human-defined assumption flagged

28. Meeting Intelligence

কাজ/উদ্দেশ্য: Discovery call-এর সিদ্ধান্ত ও follow-ups capture করা।

Screen ও sub-features: Agenda builder; participant list; notes; recording consent; summary; action extraction; CRM sync।

পূর্ণ workflow: Book meeting → attendees/calendar details sync → authorized notes/recording capture → draft summary+decisions → host verifies and approves → create CRM tasks/proposal requirements → send optional consented recap।

Data/connection: meeting, attendee, transcript_ref, summary, action_items, consent

Business rule, access ও validation: Private call recording only where lawful and approved; AI summary labeled draft Owner: SDR/Manager।

Exceptions/empty state: Provider disconnect or ambiguous speaker → manual notes

29. Sales Performance Leaderboard

কাজ/উদ্দেশ্য: Activity নয়, quality-adjusted sales outcome measure করা।

Screen ও sub-features: SDR board; conversion rates; qualified meetings; pipeline wins; response SLA; attribution of reassigned leads।

পূর্ণ workflow: Gather owned lead actions and results → exclude spam and duplicate activity → measure meeting quality, conversion and customer satisfaction → review fairness/team scope → coaching, not automatic compensation decisions।

Data/connection: user_id, leads_assigned, qualified, deals_won, response_metrics

Business rule, access ও validation: Do not reward message spamming or penalize different lead quality without context Owner: Sales Manager।

Exceptions/empty state: Reassignment attribution tracked by time and role

30. Reactivation Campaigns

কাজ/উদ্দেশ্য: Dormant eligible prospects-কে context-aware follow-up।

Screen ও sub-features: Dormant lead filter; contact preferences; suppression list; cadence templates; approval and outcome।

পূর্ণ workflow: Select leads not contacted after rule-defined period → check suppression, contact preferences, prior outcome → SDR preview suggested personalized follow-up → approved send → stop on reply/opt-out → update deal lifecycle।

Data/connection: last_contacted, lead_stage, opt_out, suppression, sequence_id

Business rule, access ও validation: No reactivation for opted-out, declined, abusive, or unsuitable recipients Owner: Sales Ops।

Exceptions/empty state: Historical contacts without lawful basis stay excluded

D4. Delivery, Client Projects & Profitability (31–40)

31. Client Onboarding Wizard

কাজ/উদ্দেশ্য: Won client handoff seamless করা।

Screen ও sub-features: Client intake form; requirements; assets; billing contact; stakeholders; kick-off checklist; access grants।

পূর্ণ workflow: Verified signed deal → draft client workspace → request intake + encrypted uploads → project manager validates scope/contacts → kickoff meeting → project board + delivery milestones activate।

Data/connection: client_id, signed_contract, onboarding_tasks, assets, team_members

Business rule, access ও validation: Do not expose project before signed agreement and authorization Owner: Delivery Manager।

Exceptions/empty state: Incomplete brief → waiting-client state, follow-up reminders

32. Scope & Change Request Manager

কাজ/উদ্দেশ্য: Unplanned work controlled costing ও approval।

Screen ও sub-features: Scope baseline; change request form; cost/time impact; approval chain; amendment; version diff।

পূর্ণ workflow: Requester submits change → PM maps to original SOW → estimates time/cost → client approves documented change → contract/invoice plan updated → tasks scheduled; rejected request stays excluded।

Data/connection: sow_version, change_request, effort_estimate, price_delta, client_approval

Business rule, access ও validation: No billable expansion without appropriate approval Owner: PM/Finance।

Exceptions/empty state: Urgent work follows authorized emergency exception with post-review

33. Project Risk Monitor

কাজ/উদ্দেশ্য: Deadline, budget, scope risk early warning।

Screen ও sub-features: Risk register; probability/impact; heat matrix; blockers; dependencies; mitigation owners; escalation।

পূর্ণ workflow: Milestone delays/time overspend/unanswered approvals detected → risk card created → PM sets likelihood, impact and actions → escalation thresholds notify leadership → verify mitigation → close risk with retrospective।

Data/connection: project, risk, likelihood, impact, blocker, action_due

Business rule, access ও validation: Heuristic risk flags are not certainty; human PM validates Owner: PM/CEO।

Exceptions/empty state: False alerts from missing timesheets get data quality label

34. Resource Capacity Planner

কাজ/উদ্দেশ্য: Team overbooking prevent এবং scheduling optimize।

Screen ও sub-features: Weekly capacity grid; skills; working hours; leave; assigned projects; forecast; what-if staffing।

পূর্ণ workflow: Staff availability and planned hours loaded → PM allocates person/project/task → overload conflict flagged → manager reschedules or finds capacity → team calendars updated।

Data/connection: member, skill, availability, allocation, leave, capacity_window

Business rule, access ও validation: Respect employee privacy and contracted working hours Owner: PM/Team Manager।

Exceptions/empty state: Missing availability → cannot assume 40 hours; flag unknown

35. Timesheet & Cost Tracker

কাজ/উদ্দেশ্য: Project actual effort ও cost measured করা।

Screen ও sub-features: Timer/manual entry; project/task selectors; hourly cost policy; approvals; corrections; utilization reporting।

পূর্ণ workflow: Employee logs task/time with note → manager reviews outliers/duplicates → approved time linked to project cost center → profitability updates → corrections recorded as revisions।

Data/connection: timesheet, task, time_range, cost_rate_ref, approver

Business rule, access ও validation: Sensitive pay rate hidden from unauthorized staff Owner: Team/Finance।

Exceptions/empty state: Overlapping entries and retroactive edits need review

36. Project Profitability Calculator

কাজ/উদ্দেশ্য: Actual margin per client/project/service।

Screen ও sub-features: Booked/paid revenue; allocated staff/vendor costs; gross margin; WIP; profitability trend; scenarios।

পূর্ণ workflow: Join signed project value and verified invoices/payments → allocate approved labor/vendor costs → calculate gross margin and cash realized separately → drilldown each allocation → PM sees permitted summary → finance reconcile monthly।

Data/connection: project_id, revenue_line, approved_costs, allocation, margin

Business rule, access ও validation: Exclude unverified promised revenue from cash figures Owner: Finance/CEO।

Exceptions/empty state: Shared overhead classification and currency rate disclosed

37. Milestone Approval Portal

কাজ/উদ্দেশ্য: Client feedback এবং acceptance traceable করা।

Screen ও sub-features: Deliverable viewer; version; feedback; approve/request change; due date; audit; comment thread।

পূর্ণ workflow: PM submits milestone version → client receives scoped portal request → previews secure artifact → Approve or Changes requested → signature/time logged → approved milestone eligible for invoice according to contract।

Data/connection: milestone, deliverable_revision, reviewer, decision, evidence

Business rule, access ও validation: Approver must be contractual authorized client contact Owner: PM/Client।

Exceptions/empty state: Link expired/multiple feedback conflicting → latest explicit decision wins

38. QA & Launch Checklist

কাজ/উদ্দেশ্য: Launch-ready evidence before live deployment।

Screen ও sub-features: Security/functional/mobile/SEO/accessibility lists; pass/fail evidence; blocker gating; launch approval।

পূর্ণ workflow: Create checklist from project type → QA executes tests with evidence → critical failures block launch → developer fixes → re-test → client/PM signs release decision → deployment reference stored।

Data/connection: checklist_template, test_case, result, screenshot, approval

Business rule, access ও validation: Critical security/privacy defect never silently waived Owner: QA/CTO/PM।

Exceptions/empty state: Failing external dependency → postpone or documented limited release

39. Client Satisfaction Tracker

কাজ/উদ্দেশ্য: Delivery quality signals capture এবং recover dissatisfied clients।

Screen ও sub-features: CSAT/NPS survey design; milestone pulse; qualitative feedback; issue routing; trend charts।

পূর্ণ workflow: After milestone/project close invite client once → collect rating and feedback with purpose → low satisfaction creates manager follow-up ticket → resolution logged → trend aggregate by service/project।

Data/connection: survey, response, project_id, score, comment, follow_up

Business rule, access ও validation: Use opt-in and avoid public testimonial use without permission Owner: Account Manager।

Exceptions/empty state: Small response sample cannot be generalized; anonymous replies protected

40. Maintenance Contract Manager

কাজ/উদ্দেশ্য: Recurring support commitments and renewals।

Screen ও sub-features: Service tiers; SLA; contract dates; tickets; usage; recurring invoices; renewal reminder; cancellation।

পূর্ণ workflow: Contract signed → support package, coverage hours and SLA configured → client tickets linked to package → utilization and SLA reported → renewal quote created before end date → authorized renewal or termination।

Data/connection: maintenance_contract, tier, hours, tickets, renewal_date, invoice

Business rule, access ও validation: No auto-charge without applicable agreement/processor mandate Owner: Support/Finance।

Exceptions/empty state: Expired contract tickets follow escalation/contract policy

D5. Infrastructure & Engineering Control (41–45)

41. Deployment History

কাজ/উদ্দেশ্য: Website/client project release impact trace করা।

Screen ও sub-features: Release list; environment; git SHA; deploy actor; build status; rollback reference; issue links।

পূর্ণ workflow: CI/CD webhook release event → normalize environment/commit/version → build/health status update → deploy incident links created → authorized operator can request rollback under policy → verify recovered version।

Data/connection: deployment_id, commit_sha, environment, artifact_ref, actor, status

Business rule, access ও validation: No arbitrary shell access from browser; prod rollback needs change-control Owner: Engineering/CTO।

Exceptions/empty state: Missing webhook → last sync warning; rollback failures escalated

42. Incident Management

কাজ/উদ্দেশ্য: Service disruptions response coordinate করা।

Screen ও sub-features: Incident board; severity; timeline; on-call; customer impact; status page; root cause; postmortem।

পূর্ণ workflow: Monitoring/employee reports issue → severity triage → incident commander assign → containment and communication → fix/verify → resolve → retrospective and preventative issue; SLA reporting linked।

Data/connection: incident_id, impact_scope, severity, timeline, mitigation, RCA

Business rule, access ও validation: Only verifiable impact/downtime communicated; sensitive details sanitized Owner: Engineering/Support।

Exceptions/empty state: Duplicate incident merged without losing original timestamps

43. Domain & SSL Renewal Tracker

কাজ/উদ্দেশ্য: Domain/SSL expiration এবং ownership risks।

Screen ও sub-features: Domain inventory; registrar; expiry; TLS certificate check; reminder rules; responsible owner; renewal evidence।

পূর্ণ workflow: Approved domain inventory created → periodic DNS/TLS checks → expiry threshold alerts → finance/owner renews via registrar → proof/reference logged → verify new expiry।

Data/connection: domain, owner, registrar, certificate_valid_to, renewal_ref

Business rule, access ও validation: Do not store registrar credentials plaintext Owner: IT/Finance।

Exceptions/empty state: Auto-renew flag not proof of renewal; verify after due

44. Technical Debt Register

কাজ/উদ্দেশ্য: Long-term maintainability prioritize করা।

Screen ও sub-features: Debt backlog; severity; affected module; risk; effort estimate; linked issues; remediation roadmap।

পূর্ণ workflow: Engineer files item with evidence → Tech Lead categorizes security/performance/maintainability → score impact vs effort → roadmap prioritization → PR/test links → verify and close after deployment।

Data/connection: debt_item, component, priority, estimate, issue_url, PR_ref

Business rule, access ও validation: Security vulnerability takes precedence and private disclosure Owner: Tech Lead।

Exceptions/empty state: Duplicated reports merged, accepted debt gets expiration review

45. Client Hosting Inventory

কাজ/উদ্দেশ্য: Client infrastructure contracts এবং expiry audit।

Screen ও sub-features: Client app/domain/environment registry; host/vendor; support ownership; backups; SSL; renewal; outage dependencies।

পূর্ণ workflow: New contracted client project → PM/engineer registers environments and assets → access references stored in vault → uptime/backups linked → renewals/ownership reviewed → offboarding transfer checklist.

Data/connection: client_id, asset_type, provider, owner, SLA, vault_reference

Business rule, access ও validation: Least privilege; do not expose client secrets or white-label identity Owner: Engineering/Account Manager।

Exceptions/empty state: Unknown ownership marked and escalated before renewal

D6. Finance, Payments & Business Intelligence (46–55)

46. Multi-Currency Finance

কাজ/উদ্দেশ্য: BDT/USD/GBP/AED cash and reporting consistent রাখা।

Screen ও sub-features: Currency selector; transaction original/converted; FX source/time; rate overrides with approval; realized gain/loss notes।

পূর্ণ workflow: Finance records transaction in original currency minor units → approved daily/reference FX rate captured → reporting currency equivalent computed → locked historical report preserves chosen methodology।

Data/connection: amount_minor, currency, fx_rate, fx_date, fx_source, conversion_method

Business rule, access ও validation: Never use today rate to silently rewrite historical cash receipt Owner: Finance।

Exceptions/empty state: FX missing → original currency shown, aggregate blocked/warned

47. Accounts Receivable Aging

কাজ/উদ্দেশ্য: Unpaid invoices prioritise করা।

Screen ও sub-features: Aging buckets (0–30/31–60/61–90/90+); reminders; disputed status; payment reconciliation; owner queue।

পূর্ণ workflow**:** Issue invoice and due date → settled amount imported/reconciled → daily unpaid balance and days overdue calculated → send approved reminder/task → dispute/credit-note adjustments → close once verified paid।

Data**/**connection**:** invoice_id, due_date, paid_amount, balance, dispute, aging_bucket

Business rule**,** access **ও** validation**:** Overdue computed by contract timezone/calendar rules Owner**:** Finance/Account Manager।

Exceptions**/**empty state**:** Part-pay, refund or disputed invoice clearly separated

48. Revenue by Service

কাজ/উদ্দেশ্য: কোন service better contribution করছে দেখা।

Screen **ও** sub**-**features**:** Service filter; contract/received splits; costs/margin; monthly trend; sector/country crosscuts।

পূর্ণ workflow**:** Map deal/invoice line items to controlled service taxonomy → reconcile payments → allocate direct costs → aggregate per-service earned/collected/margin → CEO compare trends।

Data**/**connection**:** service_id, revenue_line, payment_allocation, cost_center

Business rule**,** access **ও** validation**:** Multi-service deals allocate by actual line item or explicit method Owner**:** CEO/Finance।

Exceptions**/**empty state**:** Unallocated costs/revenue visible Unclassified

49. Commission Dispute Center

কাজ/উদ্দেশ্য: Referral payouts grievance resolve।

Screen **ও** sub**-**features**:** Dispute queue; referrer submissions; eligibility ledger; receipt evidence; notes; resolution; appeal log।

পূর্ণ workflow**:** Partner opens dispute referencing referral/payout → partner manager gathers evidence → finance compares signed rule and verified receipts → independent approver decides → update ledger by compensating entry → notify partner with safe explanation।

Data**/**connection**:** dispute, referral, agreement_version, evidence, decision, ledger_adjustment

Business rule**,** access **ও** validation**:** No direct mutation of settled ledger; preserve appeal/audit Owner**:** Partner/Finance।

Exceptions**/**empty state**:** Conflicting claims/legal hold → freeze disbursement until resolved

50. Client Lifetime Value (LTV)

কাজ/উদ্দেশ্য: Existing client contribution historical vs forecast বুঝতে।

Screen **ও** sub**-**features**:** Client revenue timeline; repeat purchases; retention; gross margin; forecast assumptions; cohort comparison।

পূর্ণ workflow**:** Verified client invoices/payments aggregated → refunds and delivery cost accounted → historical LTV displayed → separate modeled future LTV with assumptions and uncertainty → compare acquisition cost responsibly।

Data**/**connection**:** client_id, verified_revenue, costs, retention_period, model_version

Business rule**,** access **ও** validation**:** Forecast LTV cannot be called earned income Owner**:** CEO/Growth।

Exceptions**/**empty state**:** New clients sparse data → no confident future estimate

51. Revenue Concentration Risk

কাজ/উদ্দেশ্য: এক client/sector dependence alert।

Screen **ও** sub**-**features**:** Top client share; sector concentration; country/currency mix; scenario stress; alerts।

পূর্ণ workflow**:** Compute paid/contracted amounts by client/service/region separately → concentration ratio → compare threshold → show impact if account lost → management action for diversification।

Data**/**connection**:** client_id, revenue_type, segment, concentration_ratio

Business rule**,** access **ও** validation**:** Sensitive client names hidden from broad dashboards Owner**:** CEO/Finance।

Exceptions**/**empty state**:** Tiny sample/single client phase marked expected early-stage risk

52. Pricing Intelligence

কাজ/উদ্দেশ্য: Proposal estimate vs real cost learn করা।

Screen **ও** sub**-**features**:** Rate card; service packages; discount history; estimated vs actual hours; win/loss pricing review; profitability report।

পূর্ণ workflow**:** On deal proposed: versioned estimate → on project delivery: approved actual costs → compare variance → group by scope/complexity → manager adjusts rate card in new revision after approval।

Data**/**connection**:** pricebook, proposal, estimate, actual_cost, scope_variance

Business rule**,** access **ও** validation**:** No automatic public pricing changes without approval Owner**:** CEO/Sales/Finance।

Exceptions**/**empty state**:** Scope changes separated before interpreting quote errors

53. Expense Approval Workflow

কাজ/উদ্দেশ্য: Business spend authorization ও expense audit।

Screen **ও** sub**-**features**:** Expense request; receipt upload; category; project; approvers; spend policy; limits; reimbursement/payout export।

পূর্ণ workflow**:** Employee submits request+receipt → validate vendor, category, duplicate hash and budget → manager approval → finance verification → authorized payment → ledger reconciliation → archive evidence।

Data**/**connection**:** expense, vendor, receipt, budget, approval, payment

Business rule**,** access **ও** validation**:** Creator cannot self-approve above threshold; sensitive receipts restricted Owner**:** Finance/Managers।

Exceptions**/**empty state**:** Receipt missing/duplicate/overspend → exception queue

54. Financial Scenario Planner

কাজ/উদ্দেশ্য: Hiring/pricing/client conversion scenarios simulate।

Screen **ও** sub**-**features**:** Assumptions panel; scenario save/compare; break-even graph; hiring cost; pipeline sensitivity; export।

পূর্ণ workflow**:** CEO selects baseline actuals → edits hypothetical deal wins, pricing, expenses, staffing → model calculates projections with ranges → scenario stored separate from actuals → review action plan।

Data**/**connection**:** scenario_id, assumptions, baseline_period, forecast_output

Business rule**,** access **ও** validation**:** Display Scenario only on every view; cite input assumptions Owner**:** CEO/Finance।

Exceptions**/**empty state**:** Impossible assumptions and negative runway highlighted

55. Invoice and Payment Reconciliation

কাজ/উদ্দেশ্য: Invoice paid state authoritative করা।

Screen ও sub-features: Bank/provider import; unmatched transactions; match rules; split payment; credit notes; approval and audit।

পূর্ণ workflow: Payment provider/bank feed imports unique transaction → match invoice reference+amount+client+currency → finance confirms ambiguous matches → ledger posts one immutable settlement → invoice balance recomputes → receipt/referral triggers run।

Data/connection: transaction_id, bank_ref, invoice_id, allocated_minor, reconciliation_status

Business rule, access ও validation: Idempotent provider webhook; no duplicated settlement Owner: Finance।

Exceptions/empty state: Chargebacks/refunds create reversing entry; partial payments supported

D7. Referral & Partner Ecosystem Protection (56–62)

56. Unique Referral Links & Codes

কাজ/উদ্দেশ্য: Partner-specific campaign attribution retain।

Screen ও sub-features: Code generation; short link; partner portal; referral landing tracking; UTM; submitted lead list।

পূর্ণ workflow: Approved partner requests code → uniqueness + expiration/rule saved → share link → lawful visit records source code → user submits form → backend validates attribution and ties to partner → manager reviews।

Data/connection: partner_code, referral_link, campaign, submission_id, expiry

Business rule, access ও validation: Referral cookie not sole payout evidence; server verification needed Owner: Partner Manager।

Exceptions/empty state: Code reused/expired → referral held for manual review

57. Referral Fraud Detection

কাজ/উদ্দেশ্য: Duplicate/self/collusive submissions review।

Screen ও sub-features: Risk flags; duplicate graph; suspicious velocity; reviewer checklist; evidence; appeal।

পূর্ণ workflow: New referral → normalize company/contact/domain → compare existing CRM/partner records, self-match and repeated submissions → risk score with reason → human verification → eligible/rejected decision।

Data/connection: referral_id, normalized_company, duplicate_matches, risk_flag, evidence

Business rule, access ও validation: Risk flag is not automatic accusation; minimize payout info exposure Owner: Partner Manager/Finance।

Exceptions/empty state: False positive same company branches → manual override and log

58. Commission Hold & Release Rules

কাজ/উদ্দেশ্য: Eligibility before payout নিশ্চিত করা।

Screen ও sub-features: Policy builder; agreement versions; eligible receipts; hold period; refund clawback; pay approval queue; statements।

পূর্ণ workflow: Verified intro+Won deal+signed policy → receipt reconciled → eligible subtotal calculated after exclusions → hold window checks refunds/disputes → partner manager review → dual finance approval → payout + statement।

Data/connection: policy_version, percentage, cap, eligible_receipt, hold_end, payout_id

Business rule, access ও validation: Up to 15% ceiling unless executed individual policy states otherwise; no double pay Owner: Finance/Partner Lead।

Exceptions/empty state: Chargeback after payout → documented recovery/adjustment policy

59. White-Label Project Workspace

কাজ/উদ্দেশ্য: Partner brandingসহ confidential delivery।

Screen ও sub-features: Partner project workspace; branded labels; access map; milestone documents; masked contributors; client communications restrictions।

পূর্ণ workflow: Partner agreement sets white-label data boundary → staff creates project with partner client identity private → invite scoped partner team → deliverables approved through partner → publication blocked unless written consent।

Data/connection: partner_id, project, white_label_policy, shared_members, access_log

Business rule, access ও validation: Default deny client identity visibility to nonassigned teams Owner: PM/Partner Manager।

Exceptions/empty state: Accidental content cross-post trigger critical review

60. Partner Performance Tiers

কাজ/উদ্দেশ্য: Partner activation and support prioritize।

Screen ও sub-features: Tier rules; quality score; referrals sent/qualified/won; revenue; disputes; benefits; manual review।

পূর্ণ workflow: Quarterly verified outcomes aggregate → weighted quality/conversion metrics → partner manager proposes tier → approval → new benefits effective per agreement and date → partner notified।

Data/connection: partner_id, metrics_period, tier_version, eligible_benefits

Business rule, access ও validation: Tiers cannot change existing commission retroactively Owner: Partnership Manager।

Exceptions/empty state: Low data → provisional tier, not punitive downgrade

61. Partner Asset Library

কাজ/উদ্দেশ্য: Consistent approved marketing materials distribute।

Screen ও sub-features: Decks; logos; case studies; co-branded templates; version; license scope; download log।

পূর্ণ workflow: CMO uploads/approves collateral → tag market/partner model and expiry → partner sees permitted files → downloads/watermarked share → obsolete asset revoked।

Data/connection: asset_id, revision, approval, audience_scope, expiry

Business rule, access ও validation: No client logos/cases without usage rights Owner: CMO/Partner Lead।

Exceptions/empty state: Outdated claim/content → expire and notify users

62. Partner Onboarding Automation

কাজ/উদ্দেশ্য: Manual partner setup workload কমানো।

Screen ও sub-features: Onboarding checklist; agreement tasks; policy attestation; KYC/business verification as applicable; portals; training; reminders।

পূর্ণ workflow: Approved agreement event → onboarding template → gather documents/contacts → verify tax/payment channel where needed → provision scoped user → deliver assets/training → confirm first approved referral workflow।

Data/connection: partner_id, onboarding_task, agreement_status, portal_user, completion

Business rule, access ও validation: No portal access until required checks passed; store sensitive files securely Owner: Partner Ops।

Exceptions/empty state: Missing contract/verification blocks activation

D8. Security, Governance & Team Operations (63–70)

63. Multi-Factor Authentication (MFA)

কাজ/উদ্দেশ্য: Sensitive account takeover risk কমানো।

Screen ও sub-features: Authenticator/passkey setup; backup codes; recovery; trusted device; reauthentication on payout/settings।

পূর্ণ workflow: Login verifies primary auth → role requires MFA challenge → session issued with scope → high-risk action prompts step-up → lost-device recovery by approved admin process and audit।

Data/connection: mfa_factor, user_id, session, assurance_level, recovery_event

Business rule, access ও validation: Passkeys/TOTP preferred; backend enforcement mandatory Owner: All privileged users/Security।

Exceptions/empty state: Recovery abuse needs strong identity verification; no SMS-only dependency

64. Fine-Grained Permissions

কাজ/উদ্দেশ্য: Record এবং field-level least privilege।

Screen ও sub-features: Roles; policy matrix; test-as-role sandbox; team scopes; record sharing; access review; export policy।

পূর্ণ workflow: Admin defines role+permission → assign user to team/organization/project → API evaluates subject/resource/action/context → deny by default → audit authorized sensitive reads/writes।

Data/connection: permission, role, resource_scope, assignment, policy_version

Business rule, access ও validation: Frontend visibility only UX; backend authoritative Owner: Security/Super Admin।

Exceptions/empty state: Cross-tenant requests and privilege escalation return 403

65. Approval Matrix

কাজ/উদ্দেশ্য: High impact operations dual-control।

Screen ও sub-features: Approval templates; thresholds; approver chain; delegation; reminders; approval history; rejection reason।

পূর্ণ workflow: Action enters Pending approval with immutable draft/revision → system selects approvers by risk+amount+action → eligible reviewers inspect exact change → approve/reject → revalidate before execution → final receipt logged।

Data/connection: approval_policy, request, approver, decision, action_hash

Business rule, access ও validation: Creator/self-approver restriction at defined thresholds; approver cannot bypass evidence Owner: CEO/Finance/CMO।

Exceptions/empty state: Changed proposal invalidates previous approval; absent approver escalation

66. Data Retention Controls

কাজ/উদ্দেশ্য: PII minimize and secure lifecycle।

Screen ও sub-features: Retention schedule per data type; consent record; purge jobs; legal hold; backups policy; export/delete dashboard।

পূর্ণ workflow**:** Policy owner sets retention period/legal basis → scheduled job selects expired records → checks active contracts/legal hold → anonymize/delete securely → log purge evidence → backups age out under policy।

Data**/**connection**:** data_category, retention_rule, legal_hold, purge_job, consent

Business rule**,** access **ও** validation**:** Follow applicable law/contract; deletion scope includes processors Owner**:** Security/Data Steward।

Exceptions**/**empty state**:** Legal hold blocks deletion and displays documented reason

67. Customer Data Requests

কাজ/উদ্দেশ্য: Access/correction/deletion requests responsibly handle।

Screen **ও** sub**-**features**:** Request portal; identity check; data map; export/redaction; deletion approval; deadlines; tracker।

পূর্ণ workflow**:** Customer submits request → verify identity & applicable rights → locate related data across CRM/inbox/portal/connectors → review third-party confidentiality → fulfill correction/export/deletion where required → deliver secure receipt।

Data**/**connection**:** data_request, requester, verification, systems, completion_evidence

Business rule**,** access **ও** validation**:** Never release another party's message or confidential agreement blindly Owner**:** Privacy Lead।

Exceptions**/**empty state**:** Unable to verify identity → safe request for more evidence; no data leaked

68. Audit Trails

কাজ/উদ্দেশ্য: Critical action provenance এবং incident investigation।

Screen **ও** sub**-**features**:** Audit explorer; actor/action/entity; before/after references; correlation; IP metadata controls; export; tamper alerts।

পূর্ণ workflow**:** API receives authorized action → append immutable security event with actor/context and result → store separate protected log → audit team filters/correlates → incident/export with strict permissions।

Data**/**connection**:** audit_id, actor, action, resource, result, timestamp, request_id

Business rule**,** access **ও** validation**:** Avoid logging secrets, full PII or card data; logs tamper-evident Owner**:** Security/Auditor।

Exceptions**/**empty state**:** Failed actions logged; no direct user edit/delete of audit events

69. Integration Health

কাজ/উদ্দেশ্য: Connector broken হলে users বুঝবে।

Screen **ও** sub**-**features**:** Status tiles; provider/scope; last good sync; expiration; event lag; quotas; reconnection wizard; alerts।

পূর্ণ workflow**:** Background health probes OAuth/webhook sync → status computed with expected cadence → errors categorized token/rate/network/schema → owner alert → reconnect with scopes → backfill missing window with dedupe।

Data**/**connection**:** integration_id, provider, scopes, sync_cursor, last_success, error_code

Business rule**,** access **ও** validation**:** Never expose raw token to UI or logs Owner**:** Engineering/Ops।

Exceptions**/**empty state**:** Expired token means disconnected, not zero traffic

70. Operational Knowledge Base

কাজ/উদ্দেশ্য: Team SOP এবং procedures current রাখা।

Screen ও sub-features: Categories; article editor; approval; owner; version; search; linked workflow; access labels; review reminders।

পূর্ণ workflow: Process owner drafts SOP → peer review → publish revision with effective date → links to admin tasks/forms → staff acknowledges critical changes → scheduled review/retirement।

Data/connection: article, version, owner, effective_date, access_scope, acknowledgement

Business rule, access ও validation: Financial/security SOP limited to authorized roles Owner: Ops/Team Leads।

Exceptions/empty state: Stale or superseded SOP prominently flagged and archived

PART E — BUSINESS COPILOT: FULL OPERATIONAL CONTRACT

E1. Copilot screens and sub-features

UI areas: Persistent Copilot drawer, full /admin/copilot workspace, Ask box, data source selector, saved prompts/playbooks, cited answer cards, preview of proposed actions, approval request modal, execution trace, history, access warning, feedback and evaluation console.

User roles: Copilot must inherit user permissions. SDR sees only own assigned leads; CEO can request authorized summaries; Finance data accessible to permitted Finance/CEO. A user cannot prompt to bypass permissions. Vendor model cannot automatically receive all client data: minimize payload, pseudonymize when possible and govern retention/DPA.

Read workflow: (1) user question → (2) identify intent/entities/time window → (3) auth policy limits data and tools → (4) query authorized analytics/CRM/finance sources → (5) answer with numbers, dates, row counts, citations/direct drilldowns → (6) disclose missing sources and assumptions → (7) enable user to open a source record or save view.

Action workflow: (1) 'Draft follow-up to client X' → (2) fetch assigned thread/record with permissions → (3) create editable draft → (4) preview recipients, exact content, channel, risk → (5) human explicit Approve/Send → (6) trusted backend executes provider action with idempotency → (7) provider result stored → (8) audit trace and user feedback. For mass sends/publish/payout/deletion/permission change, additional approver required.

Supported query examples and actual tool boundaries:

Which channel generated most profitable clients? — Data needed: UTM attribution + signed deals + reconciled payments + allocated costs; Output: Source breakdown with attribution caveat; Allowed action: Open analytics/drilldown

Which leads are waiting for a reply? — Data needed: Inbox SLA + CRM assignment; Output: Filtered lead list and overdue duration; Allowed action: Draft replies/tasks

Why did Google clicks drop? — Data needed: GSC, recent deployments, health, seasonality; Output: Evidence-backed hypotheses, not certainty; Allowed action: Create SEO investigation

Which open deals may close? — Data needed: Pipeline, age, deal stage, past win rates; Output: Explainable weighted forecast; Allowed action: Create follow-up plan

Which deliveries are risky? — Data needed: Milestones, risks, timesheets, blockers; Output: Risk board and root causes; Allowed action: Draft escalation

Draft a proposal — Data needed: Approved pricebook + client scope + deal; Output: Edit-ready proposal draft; Allowed action: Submit for review

Compare margins — Data needed: Reconciled payment and approved costs; Output: Per-project view with currency/assumptions; Allowed action: Export permitted summary

AI guardrails: Tool allowlist, input filtering, prompt-injection defense for email/docs, explicit data provenance, human approval, per-tool rate limits, row-level permissions, trace logs with secret redaction, confidence marking, fallback if provider offline, correction feedback, model evaluations. AI never directly transfers funds or permanently deletes records.

PART F — FULL CROSS-MODULE WORKFLOWS: STEP-BY-STEP

F1. Organic Google visitor → paying software client

Step **01** — Google user clicks Appiex result → GSC counts a Search click; analytics separately records an eligible site event (counts may differ).

Step **02** — Landing page views and service/CTA actions aggregate in Analytics; capture campaign/referrer only as permitted.

Step **03** — User sends project form → server validates fields, CAPTCHA/rate rules/attachment scan; create inquiry with unique idempotency token.

Step **04** — CRM matches contact/company → classifies source as organic with confidence; inbox creates new conversation.

Step **05** — Lead routing assigns SDR with SLA; notification sends internal alert.

Step **06** — SDR opens conversation → asks required discovery questions, replies through real connected email/provider; events update.

Step **07** — Lead becomes Qualified → meeting booked and deal created, budget/currency and service mapped.

Step **08** — Proposal prepared using authorized quote template → Sales approval → real send → signed contract verification.

Step **09** — Client onboarding → project schedule → delivery → invoices → finance reconciliation.

Step **10** — Executive Dashboard reports Search clicks (GSC), site form submissions, qualified leads, won contracts and collected payments separately, with attribution confidence.

F2. Referral → sales → earned commission → payout

Step **01** — Referrer provides introduction through referral form or unique code. System records source, consent/notice and secure sensitive info.

Step **02** — Duplicate check compares prior CRM leads and existing referral claims; flags suspected conflicts.

Step **03** — Partnership/Referral manager verifies genuine prior relationship, date of introduction and eligibility.

Step **04** — Verified referral opens/links CRM lead; SDR handles sales; partner sees *high-level permitted* milestones only.

Step **05** — Deal Won does not automatically create payable commission; signed terms, eligible revenue type and verification needed.

Step **06** — Finance matches cleared client payment → commission base computed by agreed policy version, cap/exclusions/holdback applied.

Step **07** — Approval matrix verifies referrer identity, conflicts, contractual constraints; payout decision recorded.

Step **08** — Payment executed via approved finance mechanism; ledger reconciled, statement made visible; refunds/disputes create adjustments.

Step **09** — Every payout drilldown connects receipt → contract → referral → agreement → approvers.

F3. Partnership application → approved white-label delivery

Step **01** — Partner applies with White Label type → application enters Screening, owner assigned.

Step **02** — Vetting checks credibility, capabilities, client confidentiality expectations; requests missing documents if necessary.

Step **03** — Discovery call and agreement drafting define service responsibility, white-label attribution, brand policy and customer access.

Step **04** — Signed agreement and screening approved → portal membership scoped; partner training/asset library enabled.

Step **05** — Partner requests project → delivery workspace created with client-identity restrictions and partner brand settings.

Step **06** — Milestone deliverables sent to partner; partner approves or requests changes; billing and payout terms handled under contract.

Step **07** — Client identity and work remain excluded from public portfolio unless separate written permission exists.

F4. Article idea → publication → SEO observation

Step **01** — SEO identifies missing useful topic using grounded queries and target audience.

Step **02** — Content task created with purpose, target page/slug, owner, internal-link targets, evidence/citation requirements.

Step **03** — Author writes draft and attaches licensed media; autosave revision IDs.

Step **04** — SEO validates title/meta/canonical/schema/headings, link integrity, originality and factual accuracy.

Step **05** — CMO approves exact revision → schedule with timezone; scheduler receives publishing job.

Step **06** — Live CMS adapter actually publishes, returns URL/build version; route inspected for render, indexability and sitemap reference.

Step **07** — SEO monitoring waits for realistic crawl/data lag and evaluates organic traffic without asserting guaranteed ranking gains.

F5. Portfolio case study with client confidentiality

Step **01** — Internal project closes milestone → editor proposes case study.

Step **02** — Project manager verifies stack/metrics/deliverables and attaches evidence.

Step **03** — Legal/client authorization gate checks named client, screenshots, logos, quotes and results.

Step **04** — White-label restriction checked; if unsafe, only anonymized/internal case may proceed.

Step **05** — Editor prepares preview; CMO approves; live publication verified.

Step **06** — Client requests takedown → case unpublished, cache invalidation and search request process considered; incident logged.

F6. Google organic traffic suddenly drops

Step **01** — Daily metric sync detects unusual click decline; detector checks provider freshness and tracking changes.

Step **02** — System correlates crawl/index errors, deployments, redirects, canonical changes, campaign seasonality.

Step **03** — Growth owner triages as SEO issue; dev owner added for site-health defects.

Step **04** — Evidence documented; fix staged through CMS/engineering, then published with approval.

Step **05** — Monitor same comparable date windows after crawl lag; report observed results vs expectation.

F7. Late first response → escalation

Step **01** — Website form accepted; conversation enters Unassigned.

Step **02** — Lead-routings assigns SDR based on priority/service/timezone.

Step **03** — First-response SLA starts; if warning threshold reached, in-app notice fires.

Step **04** — If deadline passes with no confirmed outbound response, manager escalation created with owner and timer.

Step **05** — SDR sends legitimate reply via connector; inbox receives provider acknowledgement; SLA timer stops.

Step **06** — SLA dashboard logs delay reason and staffing recommendations; failed sends do not count as response.

F8. Inquiry → quotation → contract → project margin

Step **01** — Discovery notes and ICP lead qualification produce scoped proposal draft.

Step **02** — Quote calculates line items, delivery timeline, risk allowance and expected margin from approved rate card.

Step **03** — Sales manager verifies discount, exclusions and contract terms; approved revision goes to client.

Step **04** — Signed order triggers project baseline and cost budget; resources assigned.

Step **05** — Team logs approved work hours and vendor costs; scope requests add approved change order.

Step **06** — Finance reconciles payment receipts; project dashboard shows booked value, collected cash, actual cost and gross margin separately.

F9. Scheduled social content → external provider failure

Step **01** — Content manager approves asset, caption, channel and publish time.

Step **02** — Scheduler places job in queue; connector checks token/scopes and target asset specification.

Step **03** — API returns revoked token/rate limit → job status Failed/Retry pending, not Published.

Step **04** — Alert creates integration health ticket, and editor gets safe 'Reconnect/Reschedule' actions.

Step **05** — After restoration, idempotency check prevents duplicate post; verified post URL saved.

F10. Suspected referral fraud → commission dispute

Step **01** — Duplicate engine flags matching previously owned client/lead or conflicting referral claims.

Step **02** — Referral held; partner manager requests evidence; note preserves privacy.

Step **03** — Sales checks lead creation timestamps and acquisition provenance; Finance checks actual receipts.

Step **04** — Independent reviewer decides with reference to applicable signed policy.

Step **05** — If rejected, referrer receives neutral explanation and appeal route; no payout.

Step **06** — If accepted, eligibility recalculated with approval trace and policy snapshot.

F11. Employee leaves Appiex

Step **01** — Manager initiates approved offboarding effective date.

Step **02** — System enumerates open leads, inbox threads, CRM deals, portal memberships, scheduled publishing jobs and assigned tasks.

Step **03** — Handover to valid replacement owners; external connector shares audited.

Step **04** — Disable login/MFA sessions, API tokens, invites, external access; ensure no orphaned approvals.

Step **05** — Preserve audit records under retention policy and confirm exit checklist.

F12. Client privacy/deletion request

Step **01** — Request received and identity verified; support creates privacy case.

Step **02** — Privacy lead queries CRM, inbox, portal, analytics identification, files and external processors.

Step **03** — Legal/contractual grounds and third-party confidentiality reviewed.

Step **04** — Fulfill authorized access/correction/deletion, issue processor instructions, produce redacted secure response.

Step **05** — Append fulfillment audit and retention of minimal proof; do not leak unrelated users' records.

PART G — DATA ARCHITECTURE, EVENT MODEL & API BOUNDARIES

G1. Core entities and relationships

Identity — Entities (minimum): organizations, users, memberships, roles, permissions, sessions, MFA; Key relationship: user → authorized org/record

Acquisition — Entities (minimum): site_events, sessions_aggregated, utm_touches, consent_records, attribution_edges; Key relationship: session/touch → inquiry where permitted

SEO — Entities (minimum): gsc_properties, gsc_query_daily, gsc_page_daily, seo_issues, rank_samples, crawl_issues; Key relationship: property/page/query/day

CRM — Entities (minimum): contacts, companies, inquiries, leads, deals, stages, tasks, meetings, notes; Key relationship: inquiry → lead → deal → client

Communication — Entities (minimum): channels, conversations, messages, attachments, templates, send_attempts; Key relationship: contact → conversation → provider delivery

Network — Entities (minimum): referrals, referrers, partner_applications, partners, agreements, referral_attributions, commissions; Key relationship: referral → lead → deal → eligible payment

CMS — Entities (minimum): pages, blocks, blog_posts, projects, media, revisions, approvals, publish_jobs; Key relationship: content revision → approved site publish

Delivery — Entities (minimum): clients, projects, milestones, deliverables, change_requests, resource_allocations, timesheets, risks; Key relationship: won deal → client project → milestone

Finance — Entities (minimum): invoices, invoice_lines, payments, payment_allocations, costs, credit_notes, fx_rates, payouts; Key relationship: verified payment → cash metrics/payout eligibility

System — Entities (minimum): integrations, oauth_connections, webhook_events, jobs, automations, incidents, notifications, audit_events; Key relationship: action → job → status/audit

Entity design guidance: organization_id tenant scoping; soft-delete only where lifecycle/retention demands; immutable financial/audit logs; separate personally identifying and analytics data; optimistic concurrency (version/ETag) on editors; indexes on status/owner/created_at + source provider IDs; list APIs paginated. Relationship/link records should support multiple inquiries for one contact but not count them as multiple unique leads without declared metric policy.

G2. Required first-party event taxonomy

page_view — When emitted: Eligible page load/route navigation; Important properties: path, page_id, referrer, UTM, consent; Business interpretation: Traffic observation

cta_click — When emitted: User activates tracked CTA; Important properties: element_id, service, page_path; Business interpretation: Engagement (not inquiry)

form_start — When emitted: User begins form; Important properties: form_id, page, step; Business interpretation: Form funnel

form_validation_error — When emitted: Client-side validation fails; Important properties: field_key, error_code (no entered value); Business interpretation: Usability

inquiry_submitted — When emitted: Backend accepts valid submission; Important properties: inquiry_id, service, source; Business interpretation: Accepted inbound request

referral_submitted — When emitted: Server accepts referral; Important properties: referral_id, referrer_code; Business interpretation: Referral candidate

partner_application_submitted — When emitted: Server accepts partner application; Important properties: application_id, model; Business interpretation: Partner candidate

message_received — When emitted: Deduped inbound message stored; Important properties: conversation_id, channel; Business interpretation: Inbox workload

message_sent — When emitted: Provider send acknowledged; Important properties: message_id, delivery_state; Business interpretation: Real outbound action

lead_qualified — When emitted: Human/model-assisted verified transition; Important properties: lead_id, owner; Business interpretation: Sales qualification

meeting_booked — When emitted: Meeting persisted and provider success verified if connected; Important properties: meeting_id, channel; Business interpretation: Sales action

proposal_sent — When emitted: Actual provider accepted message; Important properties: proposal_id, revision; Business interpretation: Proposal step

contract_signed — When emitted: Signature/authorization verified; Important properties: contract_id, deal_id; Business interpretation: Contracted work

payment_reconciled — When emitted: Finance ledger commits verified settlement; Important properties: payment_id, invoice_id; Business interpretation: Collected revenue

commission_approved — When emitted: Finance/policy approval complete; Important properties: commission_id, eligible_base; Business interpretation: Liability/ready payout

content_published — When emitted: Production site confirms revision; Important properties: content_id, URL, revision; Business interpretation: Actual live content

Privacy: browser events should not contain form text, phone numbers, email addresses, payment account details, tokens, secrets or full free-text messages. Server events use idempotency + HMAC-signed webhooks where applicable.

G3. Suggested API resource boundaries

GET /api/admin/overview?from=&to=&service=&country=
GET /api/admin/analytics/traffic
GET /api/admin/analytics/events
GET /api/admin/seo/queries
GET /api/admin/seo/issues
POST /api/admin/seo/issues/:id/assign
GET /api/admin/inbox/conversations
GET /api/admin/inbox/conversations/:id/messages
POST /api/admin/inbox/conversations/:id/replies
POST /api/admin/inbox/conversations/:id/assign
GET /api/admin/leads
PATCH /api/admin/leads/:id
POST /api/admin/deals
PATCH /api/admin/deals/:id/stage
POST /api/admin/proposals/:id/submit-review
POST /api/admin/proposals/:id/send
GET /api/admin/referrals
POST /api/admin/referrals/:id/verify
POST /api/admin/referrals/:id/reject
GET /api/admin/partners/applications
POST /api/admin/partners/:id/activate
GET /api/admin/commissions
POST /api/admin/commissions/:id/approve
POST /api/admin/commissions/:id/submit-payout
GET /api/admin/content/blog
POST /api/admin/content/blog
PATCH /api/admin/content/blog/:id
POST /api/admin/content/blog/:id/submit-review
POST /api/admin/content/blog/:id/publish
GET /api/admin/content/projects
POST /api/admin/content/projects
GET /api/admin/clients/:id/projects

POST /api/admin/milestones/:id/submit-approval
GET /api/admin/finance/invoices
POST /api/admin/finance/reconciliations
GET /api/admin/integrations/health
GET /api/admin/automation/executions
GET /api/admin/audit
POST /api/admin/copilot/query
POST /api/admin/copilot/actions/:id/approve

API requirements for every endpoint: schema validation, RBAC/ABAC, input sanitization, CSRF/session controls where relevant, rate limit, pagination/filters/sort, cursor for large feeds, structured errors, request IDs, audit, idempotency for side effects, optimistic concurrency, and soft failure/retry where external providers are involved. Example response contract: {data, meta:{source, captured_at, freshness, total, permissions}, error}.

G4. External integrations (connection is optional; truthfulness is mandatory)

Google Analytics **4** — Data or action: Traffic reports, provider-defined realtime; Prerequisites: Property access/authorized API credentials; Failure UX: Disconnected/Delayed with last sync

Google Search Console — Data or action: Search queries/pages, performance; Prerequisites: Verified property + consented OAuth scopes; Failure UX: Missing property/partial dates

Site backend/forms — Data or action: Inquiry, referral, partnership submissions; Prerequisites: Server APIs/webhooks, anti-spam; Failure UX: Intake failing alert, don't lose records

Gmail / approved mail service — Data or action: Read/reply with thread identity; Prerequisites: Account scopes/consent; Failure UX: Send failure/no fake success

WhatsApp Business Platform — Data or action: Receive/send where authorized; Prerequisites: Approved business account/webhooks/templates; Failure UX: Not connected/approval required

Google Calendar — Data or action: Meetings/invitations; Prerequisites: Calendar access, correct timezone; Failure UX: Booking pending or local-only

Ad networks — Data or action: Spend and campaign state; Prerequisites: Business account permissions/API access; Failure UX: No ungrounded ROAS

n**8**n — Data or action: Workflow management/executions; Prerequisites: Scoped token and secure gateway; Failure UX: Paused/failed execution visible

GitHub / CI — Data or action: Issues, PRs, commits, releases; Prerequisites: Repo access with least scopes; Failure UX: No destructive repo actions by default

Payment/finance provider — Data or action: Settlement and invoice reconciliation; Prerequisites: Authorized verified accounts; Failure UX: Pending reconciliation not Paid

E-sign service — Data or action: Contract signing; Prerequisites: Contract template and customer consent; Failure UX: Awaiting signature not Accepted

Hosting/monitoring — Data or action: Uptime, incidents, deploys; Prerequisites: Authorized metrics/health endpoints; Failure UX: Probe failure labeled uncertainty

G5. Notification and task routing rules (sample configurable policies)

New valid web inquiry — Recipient: Routing-selected SDR + manager fallback; Internal action: Create lead/thread/SLA/task; Deadline/retry: Assign immediately; escalate if unassigned

New referral — Recipient: Partner/Referral manager; Internal action: Verification task; Deadline/retry: Based on team policy

Partner application — Recipient: Partner manager; Internal action: Screening task; Deadline/retry: Follow policy

Incoming reply to open deal — Recipient: Assigned SDR; Internal action: Reopen thread, notify; Deadline/retry: Respect business hours; SLA policy

Overdue follow-up — Recipient: Task owner → sales manager; Internal action: Alert/escalation; Deadline/retry: Dedupe; avoid alarm storm

Blog approved/scheduled — Recipient: Content publisher; Internal action: Publish job; Deadline/retry: Retry with idempotency

Publish failure — Recipient: Content owner + engineer; Internal action: Incident/repair task; Deadline/retry: Immediate critical alert if public outage

Commission ready for approval — Recipient: Finance + separate approver; Internal action: Reconcile/evidence check; Deadline/retry: Never automatic fund transfer

Domain/SSL approaching expiry — Recipient: Ops + accountable owner; Internal action: Renewal task; Deadline/retry: Verify actual renewal

Integration expired — Recipient: Integration owner; Internal action: Reconnect job; Deadline/retry: Last-success timestamp maintained

PART H — IMPLEMENTATION BACKLOG, PHASING & ACCEPTANCE

H1. Suggested phased build (do not build 90 modules simultaneously)

R0: foundation — Core deliverables: Auth/MFA, org/roles, app shell, audit, entity IDs, intake APIs, demo-safe adapters; Dependencies: Identity, security, deployment; Definition of Done: No sensitive API accessible by wrong role; no lost form intake

R1: acquisition & sales — Core deliverables: Overview, GA4/GSC read, website tracking, inbox, CRM, referral/partner intake, notifications; Dependencies: Connectors, domain tracking, normalized schema; Definition of Done: Valid lead goes form→queue→reply→deal; metrics sourced

R2: CMS & sales ops — Core deliverables: Blog, project, website CMS, approvals, scheduling, campaigns, proposals, calendars; Dependencies: Public site publishing API, email/calendar; Definition of Done: Publish verified; proposal approved and provider sent

R3: finance & delivery — Core deliverables: Signed contracts, client portal, milestones, invoices, reconciliation, commission ledger, resource/timesheets; Dependencies: Financial policy, client agreements; Definition of Done: Accounting and payout states reconcile; external scopes safe

R4: growth intelligence — Core deliverables: Experiments, session analytics with consent, advanced SEO, attribution, forecasts; Dependencies: Quality historical data, privacy controls; Definition of Done: Results explainable, no fabricated inference

R5: AI & automation — Core deliverables: Copilot, AI scoring/coach, n8n, health/infrastructure control; Dependencies: Stable permissions, job audit, evaluation; Definition of Done: Human approval for all high-impact external actions

H2. Universal acceptance criteria applied to every feature

Step 01 — Screen: Fully responsive desktop/tablet/mobile; keyboard accessible and good contrast; loaders/errors/empty states.

Step 02 — Action: Every button does something truthful: operate API or clearly labeled local demo; no dead UI.

Step 03 — Data: Validation, source, freshness, filters, pagination and states are correct.

Step 04 — Permissions: Server-side role AND resource ownership checks; unauthorized access returns proper error.

Step 05 — Audit: Sensitive change, approval, payout, publication and permission changes recorded.

Step 06 — Safety: Secrets and personal data never leak in logs, URL parameters or analytics events.

Step 07 — Reliability: Retries and duplicate webhooks do not double-send, double-create or double-pay.

Step 08 — Integration: Revoked/expired connections show explicit disconnected state; no phantom metrics.

Step 09 — Reconciliation: Aggregates trace to actual records and explain exclusions.

Step **10 —** Outcome**:** End-to-end test reproduces intended workflows and adverse scenarios, not just screenshots.

H3. Minimum practical E2E test suite

T**01** — Scenario**:** Valid contact form and duplicate retry; Expected verifiable result**:** Exactly one accepted inquiry + owned queue item

T**02** — Scenario**:** Spam referral attempt; Expected verifiable result**:** Quarantined/flagged; not qualified or payable

T**03** — Scenario**:** SDR reply via disconnected email channel; Expected verifiable result**:** Failed/pending; never shown sent

T**04** — Scenario**:** Lead qualified and won; Expected verifiable result**:** Linked deal/client with preserved acquisition data

T**05** — Scenario**:** Quote revised after approval; Expected verifiable result**:** Previous approval invalidated; new approval required

T**06** — Scenario**:** Editor publishes blog with working adapter; Expected verifiable result**:** URL/build revision verified; audit recorded

T**07** — Scenario**:** Demo-only project CMS publish; Expected verifiable result**:** Clearly simulated; public site unchanged

T**08** — Scenario**:** White-label case study without permission; Expected verifiable result**:** Publication blocked

T**09** — Scenario**:** Partial invoice payment; Expected verifiable result**:** Balance/commission eligibility calculated by terms

T**10** — Scenario**:** Duplicate payment webhook; Expected verifiable result**:** Exactly one ledger settlement

T**11** — Scenario**:** Commission approve and payout retry; Expected verifiable result**:** No duplicate payout; independent approval verified

T**12** — Scenario**:** SDR requests another team's finance data; Expected verifiable result**:** API returns 403; UI permission denied

T**13** — Scenario**:** Partner opens another partner's referral; Expected verifiable result**:** Denied; no cross-tenant leak

T**14** — Scenario**:** Google Search Console disconnected; Expected verifiable result**:** Not connected, no fake zero/position

T**15** — Scenario**:** Session replay without consent; Expected verifiable result**:** No recording; aggregated alternative only

T**16** — Scenario**:** Uptime probe false positive; Expected verifiable result**:** Incident triage can mark probe error with evidence

T**17** — Scenario**:** Copilot prompted to secretly pay commission; Expected verifiable result**:** Action denied; approval required

T**18** — Scenario**:** Scheduler provider token expired; Expected verifiable result**:** Failed/reconnect UI; no published status

T**19** — Scenario**:** Employee offboarding; Expected verifiable result**:** Sessions revoked; assignments transferred

T**20** — Scenario**:** Customer data request; Expected verifiable result**:** Verified and fulfilled with correct privacy protections

H4. Product and design implementation standards

Reuse existing Appiex branding (actual available brand files first); fallback graphite/dark surfaces, restrained red only if brand assets confirm or no official tokens exist.

Maintain normalized route IDs; all metric cards have meaningful definitions, timestamp and drilldowns.

Use typed API clients and separate backend services by domain; consider event-driven outbox/jobs for side effects. Database schema changes versioned and tested.

External platform accounts and official API terms must be verified before implementation; this document describes proposed capabilities, not proof every API supports every action.

Security model: HTTPS, encrypted sensitive fields, MFA, least privilege, secret manager, malware scan, rate limiting, approval chain, disaster recovery, availability monitoring and backup verification.

SEO system should guide useful content creation and technical quality; no guaranteed top ranking and no mass low-quality keyword stuffing.

H5. Development handoff instruction

Implement APPIEX ADMIN OS v**2.0** by treating this document as the complete functional workflow and acceptance source of truth. Before building, inspect existing repo, site forms, schema, and available integrations. Map current functionality to each feature; preserve working code. Implement foundation and core intake/inbox/CRM first, then content workflows, finance/partners, advanced measurement, and finally AI automation. For each feature deliver the actual pages, all sub-feature controls, typed APIs/state transitions, validation,

permission checks, audit trails, test fixtures, E2E scenarios and clearly marked disconnected/demo states. Do not confuse UI prototype with completed backend integration. Never claim public publication, message delivery, payment, SEO ranking or real traffic without authentic evidence.

PART I — FRONTEND IMPLEMENTATION: COMPLETE ROUTE-TO-SCREEN MAP

প্রতিটি route একটি working page হবে। Contextual drawer/modal/confirmation, loading, retry, empty state, permission denied, disconnected provider এবং responsive view থাকা বাধ্যতামূলক। :id হলো authorized dynamic resource identifier। Route-গুলো প্রস্তাবিত; existing repository-র convention অনুযায়ী নাম বদলানো যেতে পারে।

Every primary action must produce a real documented backend change or visible Demo-only state. List filters persist in query parameters without leaking PII; detail pages include ownership, notes, linked records and audit history.

I. Command & Executive

Screen 01 — /admin

Visible elements: All-business KPI, attribution, pending tasks, integration freshness।

User actions: Date filters; drilldown; assign urgent task; export authorized।

Data/connector: Dashboard aggregation। Access: CEO, leadership।

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs।

Screen 02 — /admin/realtime

Visible elements: Realtime visitors, active pages, incoming events, source health।

User actions: Page drilldown; pin event; alert owner।

Data/connector: GA4 realtime/event stream। Access: Growth।

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs।

Screen 03 — /admin/notifications

Visible elements: Unread/critical/snoozed alerts, delivery timeline।

User actions: Open related record; mark read; snooze; edit preferences।

Data/connector: Notification service। Access: All staff।

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs।

Screen 04 — /admin/reports

Visible elements: Saved KPI reports, schedules, previous exports।

User actions: Create report; schedule; download; revoke shared link।

Data/connector: Report job engine। Access: Executive/Analyst।

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 05 — /admin/copilot

Visible elements: Grounded Q&A, supporting records, pending AI actions |

User actions: Ask; inspect sources; edit draft; request approval |

Data/connector: Policy-scoped query and action services | Access: All, access-scoped |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

I. Growth & SEO

Screen 06 — /admin/analytics

Visible elements: Visits, sessions, engagement, device/country/channel |

User actions: Filter cohort; compare; open source; save view |

Data/connector: GA4, first-party events | Access: Growth |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 07 — /admin/analytics/pages

Visible elements: Top pages, CTA clicks, bounce proxies, funnels |

User actions: Inspect path; open page; create optimization ticket |

Data/connector: Analytics events, CMS | Access: Growth |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 08 — /admin/analytics/events

Visible elements: Event explorer, taxonomy validity, ingestion errors |

User actions: Search events; inspect schemas; flag broken tracking |

Data/connector: Event pipeline | Access: Analyst/Engineering |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 09 — /admin/analytics/journeys

Visible elements: Privacy-filtered journey sequences |

User actions: Select cohort; inspect masked journey; retention controls |

Data/connector: Consent + event pipeline | Access: Authorized Analyst |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 10 — /admin/analytics/heatmaps

Visible elements: Aggregated scroll and click attention per page |

User actions: Choose viewport/page version; compare; create experiment |

Data/connector: Consent-safe tracking | Access: UX/Growth |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 11 — /admin/analytics/forms

Visible elements: Form start, errors, completion, abandonment |

User actions: Inspect failed step; assign bug; compare variants |

Data/connector: First-party frontend and backend events | Access: Growth/Product |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 12 — /admin/analytics/audience

Visible elements: New/returning, geo, device, campaign cohorts |

User actions: Define/save segment; compare conversion |

Data/connector: GA4/provider aggregates | Access: Growth |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 13 — /admin/analytics/attribution

Visible elements: Source journey to lead/deal/paid invoice |

User actions: Choose model; see unknown bucket; drill into receipt |

Data/connector: CRM+UTM+finance ledger | Access: Growth/Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 14 — /admin/seo

Visible elements: Search clicks/impressions/CTR/position, issue queue |

User actions: Filter GSC property; open opportunity; create SEO task |

Data/connector: Google Search Console | Access: SEO/CMO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 15 — /admin/seo/queries

Visible elements: Keyword-query report, page mapping, search intent |

User actions: Inspect query; map page; add monitored keyword |

Data/connector: GSC query samples | Access: SEO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 16 — /admin/seo/pages

Visible elements: Organic landing URL performance, decay |

User actions: Open URL; compare date; draft content refresh |

Data/connector: GSC URL aggregates | Access: SEO/Content |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 17 — /admin/seo/technical

Visible elements: Index/robots/sitemaps, redirects, CWV, schema |

User actions: Open error; assign engineering issue; verify resolution |

Data/connector: Crawler, PageSpeed, GSC | Access: SEO/Engineering |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 18 — /admin/seo/opportunities

Visible elements: CTR/decay/cannibalization/link opportunities |

User actions: Evaluate evidence; priority; push to task/CMS |

Data/connector: SEO analytics and page graph | Access: SEO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 19 — /admin/seo/ai-visibility

Visible elements: Documented AI answer sampling and mentions |

User actions: Run allowed sample; inspect snapshot; export caveated trend |

Data/connector: Approved sampling sources | Access: SEO Research |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 20 — /admin/seo/backlinks

Visible elements: Relevant prospective sites and earned references |

User actions: Qualify; track pitch; verify link |

Data/connector: Permitted public evidence | Access: SEO outreach |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 21 — /admin/campaigns

Visible elements: Cross-channel campaigns, spend, UTMs, lead count |

User actions: Create campaign; build UTM; link creative; approve |

Data/connector: Ad connectors + CRM | Access: CMO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 22 — /admin/experiments

Visible elements: Hypotheses, variants, active experiments, outcomes |

User actions: Draft; QA; approve split; pause; rollout |

Data/connector: Experiment SDK and events | Access: CMO/Product |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

I. Sales & Communications

Screen 23 — /admin/inbox

Visible elements: Conversation list, assignment, SLA, thread composer |

User actions: Reply; note; reassign; link lead; resolve; search |

Data/connector: Email/form/approved WhatsApp APIs | Access: SDR/Support |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 24 — /admin/contacts

Visible elements: People, companies, duplicate candidates, activity |

User actions: Merge verified; enrich; link to lead; export approved |

Data/connector: CRM | Access: Sales |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 25 — /admin/leads

Visible elements: Lead source, qualification, score, owner, next action |

User actions: Qualify; disqualify with reason; assign; create deal |

Data/connector: CRM and intake | Access: SDR |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 26 — /admin/pipeline

Visible elements: Deal kanban, stage, value, owner, close forecast |

User actions: Move deal; schedule next action; open proposal |

Data/connector: CRM | Access: Sales |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 27 — /admin/tasks

Visible elements: My tasks, overdue, team SLA, reminders |

User actions: Complete; reassign; reschedule; bulk plan |

Data/connector: Task service | Access: Team/Manager |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 28 — /admin/meetings

Visible elements: Calendar availability, bookings, outcomes, notes |

User actions: Book; reschedule; summarize; create follow-up |

Data/connector: Calendar + CRM | Access: Sales |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 29 — /admin/proposals

Visible elements: Templates, drafts, discounts, approvals, sent/accepted |

User actions: Build PDF; request approval; send; revise |

Data/connector: Proposal service/email/e-sign | Access: Sales/Management |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 30 — /admin/sales/sequences

Visible elements: Step templates, enrollments, replies, opt-outs |

User actions: Create; preview; approve; enroll; pause |

Data/connector: Mail connector + suppression | Access: Sales Ops |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 31 — /admin/sales/coaching

Visible elements: Call summaries, objection library, suggestions |

User actions: Inspect cited advice; accept/reject; draft follow-up |

Data/connector: Consent-aware AI tools | Access: SDR/Manager |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 32 — /admin/sales/forecast

Visible elements: Pipeline predictions, assumptions, actuals |

User actions: Compare scenarios; export assumptions |

Data/connector: CRM history | Access: Sales Manager/CEO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

I. Network & Relationships

Screen 33 — /admin/referrals

Visible elements: All submissions, risk flags, duplicates, eligibility |

User actions: Review; request evidence; verify/reject; link lead |

Data/connector: Site referral API + CRM | Access: Referral Manager |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 34 — /admin/referrals/:id

Visible elements: Referrer, intro evidence, linked deal, receipts |

User actions: Audit eligibility; open dispute; notify referrer |

Data/connector: Referral ledger | Access: Partner/Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 35 — /admin/partners/applications

Visible elements: White Label/Affiliate/Collaboration applications |

User actions: Screen; schedule call; reject; request contract |

Data/connector: Partner form API | Access: Partner Manager |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 36 — /admin/partners

Visible elements: Active partner list, model, tier, origin value |

User actions: Open; adjust approved tier; suspend; renew |

Data/connector: Partner CRM | Access: Partnerships |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 37 — /admin/partners/:id

Visible elements: Agreement, shared projects, attributed leads, status |

User actions: Onboard; invite scoped user; set restrictions |

Data/connector: Partner CRM, portals | Access: Partner Manager |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 38 — /admin/commissions

Visible elements: Accrued/held/approved/paid balances and disputes |

User actions: Inspect receipt; approve; send payout for execution |

Data/connector: Immutable finance/referral ledger | Access: Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 39 — /admin/partner-assets

Visible elements: Approved decks, logos, white-label collateral |

User actions: Upload; review; grant/revoke access |

Data/connector: Media + access policy | Access: CMO/Partner Manager |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

I. Website & Content Studio

Screen 40 — /admin/content/pages

Visible elements: Page sections, navigation and homepage blocks |

User actions: Edit copy/media; preview; request approval; rollback |

Data/connector: Public website CMS/publishing adapter | Access: Editor/CMO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 41 — /admin/content/services

Visible elements: Five services, capabilities, CTA and SEO |

User actions: Create/edit service; reorder; preview; publish |

Data/connector: Service CMS | Access: Content/CMO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 42 — /admin/content/blog

Visible elements: Blogs, statuses, tags, scheduled posts, results |

User actions: New; edit; duplicate; submit; schedule; archive |

Data/connector: Blog CMS | Access: Content Editor |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 43 — /admin/content/blog/new

Visible elements: Title, slug, body, SEO/meta, author, media |

User actions: Autosave; preview; validate; submit approval |

Data/connector: Blog CMS | Access: Content Editor |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 44 — /admin/content/projects

Visible elements: Case studies, categories, public/demo status |

User actions: New; select media; verify permission; approve |

Data/connector: Portfolio CMS | Access: Delivery/Content |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 45 — /admin/content/projects/new

Visible elements: Editor, evidence, tech stack, gallery, consent |

User actions: Save draft; open client approval; preview; publish |

Data/connector: Portfolio CMS | Access: PM/CMO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 46 — /admin/content/media

Visible elements: Images/video/documents, usage rights, alt texts |

User actions: Upload; crop; tag; deprecate; check rights |

Data/connector: Object storage + media CMS | Access: Content |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 47 — /admin/content/calendar

Visible elements: Editorial calendar, tasks, review and schedule |

User actions: Drag schedule; assign; approve; publish job |

Data/connector: Content engine + channel adapters | Access: CMO/Content |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 48 — /admin/content/testimonials

Visible elements: Testimonials, evidence, permission and expiry |

User actions: Review claim; obtain consent; publish/unpublish |

Data/connector: CMS + signed permission | Access: CMO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 49 — /admin/content/redirects

Visible elements: URL rules, links, canonical and redirects |

User actions: Propose; detect collisions; test; deploy |

Data/connector: CMS + public routing layer | Access: SEO/Engineering |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

I. Client Delivery, Finance & Engineering

Screen 50 — /admin/clients

Visible elements: Client accounts, related projects, billing and contacts |

User actions: Create onboarding; assign account manager; view history |

Data/connector: CRM and agreements | Access: Account Manager |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 51 — /admin/delivery/projects

Visible elements: Active projects, risk, resources, milestones |

User actions: Create milestone; assign staff; flag risk; launch QA |

Data/connector: Delivery service | Access: PM |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 52 — /admin/delivery/milestones

Visible elements: Deliverables, review/acceptance versions |

User actions: Submit to client; request approval; amend |

Data/connector: Portal + delivery | Access: PM/Client |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 53 — /admin/delivery/resources

Visible elements: Capacity, allocations, leave and skills |

User actions: Allocate; resolve overload; export team plan |

Data/connector: Team & delivery | Access: PM |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 54 — /admin/delivery/timesheets

Visible elements: Time entries, cost centers, approvals |

User actions: Log; review; correct; approve |

Data/connector: Timesheet/finance | Access: Staff/Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 55 — /admin/finance

Visible elements: Paid/booked/invoiced metrics, margin, balances |

User actions: Drill down; switch currency; export approved |

Data/connector: Financial ledger | Access: Finance/CEO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 56 — /admin/finance/invoices

Visible elements: Invoices, AR age, amounts, due dates |

User actions: Issue; send approved; record credit; reconcile |

Data/connector: Invoice/finance | Access: Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 57 — /admin/finance/payments

Visible elements: Bank/provider entries, settlement matching |

User actions: Match; verify; reconcile; reverse with receipt |

Data/connector: Payments integration | Access: Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 58 — /admin/finance/expenses

Visible elements: Expenses, receipt proofs, budget controls |

User actions: Submit; approve; reject; reimburse |

Data/connector: Expense service | Access: Staff/Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 59 — /admin/finance/scenarios

Visible elements: What-if cash flow, staffing, price assumptions |

User actions: Create scenario; compare; share internal |

Data/connector: Forecast service | Access: CEO/Finance |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 60 — /admin/support

Visible elements: Client tickets, status, priority, SLA |

User actions: Assign; reply; escalate; resolve |

Data/connector: Support system | Access: Support |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 61 — /admin/engineering/releases

Visible elements: Build/deploy timeline, environments, rollback ref |

User actions: Inspect release; open incident; request rollback |

Data/connector: CI/CD connectors | Access: Engineering |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 62 — /admin/engineering/incidents

Visible elements: Outages, impacted assets, actions and timeline |

User actions: Acknowledge; update; resolve; root cause |

Data/connector: Monitoring + tickets | Access: On-call/CTO |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 63 — /admin/engineering/assets

Visible elements: Domain, SSL, hosting client inventory |

User actions: Edit owner; check expiry; create renewal task |

Data/connector: Asset registry | Access: Engineering/Ops |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 64 — /admin/engineering/debt

Visible elements: Technical debt risk and remediation issues |

User actions: Score; link PR; prioritize; verify close |

Data/connector: GitHub/issues registry | Access: Tech Lead |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

I. System, Privacy & Operations

Screen 65 — /admin/team

Visible elements: Users, invitations, roles, workload, access |

User actions: Invite; assign; deactivate; initiate offboarding |

Data/connector: Identity service | Access: Super Admin/Team Lead |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 66 — /admin/permissions

Visible elements: Role, data scope and action approval rules |

User actions: Edit policy; preview denied access; review |

Data/connector: Policy engine | Access: Security |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 67 — /admin/automations

Visible elements: n8n workflows, execution health, retries |

User actions: Test; enable; pause; inspect failed run |

Data/connector: n8n + orchestration | Access: Operations |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 68 — /admin/integrations

Visible elements: Connected accounts, scopes, expiry, lag |

User actions: Connect; refresh; disconnect; run sync |

Data/connector: Integration gateway | Access: Super Admin/Ops |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 69 — /admin/audit

Visible elements: Actor, record, action, outcomes and request IDs |

User actions: Filter; inspect; export authorized logs |

Data/connector: Audit append-only log | Access: Auditor/Security |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 70 — /admin/privacy

Visible elements: Retention, consent, data access/deletion requests |

User actions: Verify request; approve fulfillment; export audit |

Data/connector: Privacy services | Access: Data Steward |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 71 — /admin/approvals

Visible elements: Content, contract, payout and sensitive changes |

User actions: Compare revision; approve/reject; delegate |

Data/connector: Approval engine | Access: Authorized Approver |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 72 — /admin/knowledge

Visible elements: SOPs, runbooks, templates and revisions |

User actions: Write; review; publish; retire |

Data/connector: Knowledge repository | Access: Operations |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

Screen 73 — /admin/settings

Visible elements: Brand/currency/timezone/notification and API health |

User actions: Update scoped settings; test; audit |

Data/connector: Org configuration service | Access: Super Admin |

Interaction pattern: List → filter/search → select record → contextual detail → validate → perform allowed action → show authoritative state/confirmation → append audit and refresh dependent KPIs |

I. Screen-level UI state contract

- Initial: skeleton + date/source placeholders; no metrics until data resolves.
- Empty: explain no matching records, suggest clearing filters, never fabricate rows.
- Disconnected: exact required connector and permission shown with safe reconnect path.
- Pending approval: immutable proposed change + difference viewer + approval route.

- Error: sanitized error, correlation/request ID, retry if safe; do not silently mark sent/paid/published.
- Success: provider-backed record ID/link, source timestamp, and audit entry.
- Permission denied: restricted content redacted server-side, not only hidden in DOM.
- Mobile: usable filters and row actions in stacked cards/drawers, no horizontal overflow of forms.

End of APPIEX Admin OS Complete Feature & Sub-Feature Workflow Blueprint — v2.0.